

POKEMON GOLD HACK STUDY SNAPSHOT

Boss AI Teaching Heuristics

A print-ready reading copy of the active boss AI teaching notebook, captured without modifying the working Markdown file.

SNAPSHOT TAKEN

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SOURCE

C:\Users\lolno\Downloads\pokemon gold hack\docs\boss_ai_teaching_heuristics.md

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2,942 LINES	87 HEADINGS	11 TABLES	167,517 BYTES
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MAJOR SECTIONS

- Workflow
- Synthetic Data Rules
- Pokemon Singles Study Notebook
- Current Heuristics
- Answered Teaching Notes
- Current Scorer Gap Audit

Boss AI Teaching Heuristics

Purpose: preserve the user's boss-battle judgment as readable rules before turning it into synthetic examples, debugger weights, or ROM scoring changes.

This file is the working sheet for the hybrid teaching loop. A rule should not be promoted into ROM behavior just because it appears here; it needs matching fixtures, passing audits, and a clear implementation target.

Workflow

1. Ask the user a concrete public-info boss state from the training lab.
2. Record the answer as a rule, exception, or open question here.
3. Save the matching preference or trajectory row in the lab data.
4. Re-run reports to check for conflicts, stale source rows, and coverage gaps.
5. Generate synthetic examples only from stable, high-confidence rules.
6. Keep user-authored examples weighted above synthetic examples.
7. Promote a rule to ROM scoring only when the behavior is clear and testable.

Synthetic Data Rules

- Synthetic examples are allowed for obvious cases, not taste calls.
- Synthetic rows must stay marked as synthetic when generated.
- Do not generate synthetic labels from a rule with unresolved exceptions.
- Do not use synthetic data to override a direct user answer.
- Prefer small batches that can be spot-checked before larger batches.

Pokemon Singles Study Notebook

Status: active learning. Do not treat this section as mastery proof yet. It is the working notebook for learning the game well enough to answer Gym Leader Lab questions without sounding like a generic policy summary.

Operational Mastery Target

I am not "done" when I can name a good move in a vacuum. The target is to play or analyze a long singles battle by tracking:

- the current damage race;
- both players' likely win conditions;
- which Pokemon must be preserved for later;
- which opposing answers have been revealed, weakened, slept, paralyzed, or

removed;

- what happens after taking a KO;
- what happens if a setup, sleep, recovery, or switch line fails;
- PP, recovery, hazards, status, and forced-switch consequences over many

turns.

Before choosing a move, ask these questions in order:

1. Can I win or prevent losing immediately?
2. Does a direct KO create a worse board for me on the next forced switch?
3. Which side improves if both players make the obvious safe play?
4. What is the worst plausible punish, and can my team absorb it?
5. Does setup change the number of turns needed to win, or is it just greed?
6. Is status buying a concrete future turn, denying recovery, or preserving a

resource?

1. Is switching preserving something that matters, or just giving away tempo?
2. After this exchange, what is my remaining route to win?

Research Anchors

Current web research agrees with the user-taught direction:

- Smogon's beginner guide frames good play as minimizing risk while maximizing

reward each turn, then checking the long-term effect of a move: what comes in after a KO, and how a sweeper gets a setup opportunity.

- Smogon's risk/reward guide separates the likely opposing move from the

worst-case move, and says conservative play is strongest when the matchup is already favorable while riskier play may be needed from a bad matchup.

- Smogon's long-term thinking guide emphasizes that battles are not one-turn

puzzles; preserve sweepers until counters are known or weakened, track PP, and use recovery turns as chances to create pressure.

- GSC Spikes writing treats status as long-term pressure, not just immediate

disruption: status can deny spin, force recovery, and support future entry or setup lines, but it needs patience and a plan against cleric/recovery answers.

- GSC-specific writing emphasizes that normal GSC is not won by autopilot

switching. You need a plan and backup plan, look many turns ahead, exploit switching patterns, and avoid relying on Snorlax as the only win condition.

- Win-condition writing frames a win condition as an endgame route, not just a

strong Pokemon: identify what threatens you, remove or weaken those answers, and update the plan as information changes.

- GSC mechanics notes matter for this hack's base: one-layer Spikes, Sleep Talk

can call Rest, and GSC Roar/Whirlwind have generation-specific behavior.

- Borat's GSC guide is a useful warning against shallow one-turn thinking:

kills are worked for, switches must be punished, and useful planning can run many turns ahead rather than only predicting the next click.

- GSC threat writing treats Snorlax as the central threat and warns that no

single answer covers all sets. Preserve multiple Snorlax answers or a real emergency route instead of assuming one check solves the matchup.

- GSC Explosion writing treats Explosion as wallbreaking, emergency defense,

free-turn creation, or endgame simplification depending on context. It is a future-route trade, not just a high-damage move.

- GSC Misdreavus analysis supports the trap-clock lesson from the scratch

battles: Ghost typing, spinblocking, Snorlax checking, Mean Look, Perish Song, and Pursuit vulnerability form a whole route map. The answer is not "attack the Ghost"; the answer is to know whether phazing, Pursuit, sleep state, or immediate KO pressure can change the countdown.

- GSC SleepTrap discussion reinforces that sleep plus trapping can bypass the

normal "fodder something to sleep, then answer it" pattern. Any sleep/trap line in the romhack must therefore be treated as a mechanics and ruleset question, not just a standard status heuristic.

Sources:

- <https://www.smogon.com/articles/getting-started>
- https://www.smogon.com/resources/beginner/bw_risk_reward
- https://www.smogon.com/rs/articles/long_term_thinking
- https://www.smogon.com/gs/articles/gsc_spikes

- <https://www.smogon.com/smog/issue28/gsc>
- https://www.smogon.com/gs/articles/gsc_guide_part1
- https://www.smogon.com/gs/articles/gsc_threats
- https://www.smogon.com/gs/articles/guide_to_explosion
- <https://www.smogon.com/forums/threads/misdreavus-ou-revamp-done.3643258/>
- <https://www.smogon.com/forums/threads/sleep-trapping.3617871/>
- <https://www.smogon.com/forums/threads/knowning-how-to-find-your-win-condition.3474271/>
- <https://www.smogon.com/forums/threads/gsc-mechanics.3542417/>
- <https://github.com/smogon/pokemon-showdown>
- <https://www.npmjs.com/package/%40pkmn/sim>

Evidence Tiers

Keep these separate:

- General strategy source: Smogon articles, Showdown-style simulations, and

human review. Use this to learn concepts like tempo, win conditions, status pressure, setup windows, and long-term planning.

- Romhack source evidence: source-derived fixtures, trainer rosters, move data,

type charts, and threat availability reports. Use this before answering a Gym Leader Lab state.

- Romhack runtime evidence: damage debugger, PyBoy, trace ROM captures, and

live battle states. Use this before claiming exact damage, exact legality, AI behavior, or source-level correctness.

Do not promote a Showdown-sim result to a boss heuristic unless the same idea still makes sense against the romhack fixture data and known mechanics.

Romhack-Specific Simulation Path

Easiest useful path for Gym Leader Lab work:

1. Start from `tools/boss_ai_preference/fixtures/boss_ai_preference_fixtures.json`

so teams, known moves, public priors, boss rosters, and field context come from this repo instead of a generic simulator.

1. Use `tools.damage_debugger.matchup` or the preference lab's damage-estimate

helpers for concrete damage questions before trusting type or KO intuition.

1. Build a small branch evaluator for lab-shaped questions:

sleep/setup/attack, switch/preserve, recover/attack, scout/commit, and lock-in/cash-out.

1. Use PyBoy or trace ROM only when exact battle-engine behavior matters.

Full Pokemon Showdown parity is possible but should be treated as a separate project, not a quick prerequisite. Do not edit installed `node_modules` directly. If a playable sandbox becomes worth it, make a local Showdown mod or fork and port mechanics incrementally from the repo docs and data. The first version should cover only the mechanics the current lab questions need.

Transfer Model

Learning normal GSC is still worthwhile. Treat it as the base language for:

- long-term win-condition planning;
- Rest, Sleep Talk, status, and recovery cycles;
- using sleep or paralysis to create setup turns;
- preserving key Pokemon instead of winning only the current turn;
- switching, double-switching, phazing, and punishing predictable patterns;
- turning Spikes/status chip into future KO ranges.

The romhack is the dialect layer. Before answering a lab question, check the local differences that can change the conclusion. The largest current caution is three-layer Spikes, because it makes repeated switching much more expensive than vanilla GSC. Type passives, late-gen item behavior, and trainer-specific rosters are also enough to change damage races and switch value.

GSC Role Map Seed

Use roles to decide what can be spent. A Pokemon is "low HP" only after its job is no longer needed; otherwise it may still be the piece holding the game together.

Role	Common GSC examples	Why it matters	Preservation question
Central win route / defensive hub	Snorlax	Snorlax can attack, absorb special pressure, Rest, Curse, spread paralysis, or boom depending on set. No one answer covers every set.	What stops opposing Snorlax if this is damaged, slept, boomed, or forced to Rest?
Spiker	Cloyster, Forretress	Spikes turn switching, phazing, Rest cycles, and doubles into progress. In this romhack, 3 layers can make the role even more defining.	Can the setter place and keep hazards, or is it being spent before the hazard game starts?
Spinner / hazard control	Starmie, Forretress, Cloyster	Removing Spikes changes every future switch and can rescue a defensive gameplan.	If this dies or is statused, can the team still tolerate hazard pressure?
Spinblocker / anti-removal	Misdreavus, Gengar	Keeps Spikes down and can force awkward Normal/Ghost coverage decisions.	Is blocking spin worth the HP/status risk in this specific hazard state?
Phazer / anti-setup	Skarmory, Steelix, Raikou, Suicune	Roar/Whirlwind stops setup and turns Spikes into repeated damage. Gen 2 phazing has special priority behavior, so speed order matters.	If this phazer is lost, what stops Curse, Belly Drum, Growth, or Baton Pass routes?
Electric pressure / bulky special check	Raikou, Zapdos	Forces Water/Flying pressure, absorbs some special routes, and often participates in RestTalk cycles.	Can this afford Rest now, or will sleep turns donate tempo?
Explosion / sacrifice route	Cloyster, Exeggutor, Gengar, Steelix, Forretress, Snorlax	Explosion can remove counters, stop a sweep, create a free entry, or simplify a won endgame.	What future route does the trade buy, and what defensive job disappears?
Sleep or paralysis creator	Exeggutor, Gengar, Jynx, Nidoking, Lovely Kiss Snorlax	Status creates turns, denies recovery, and shapes switch patterns.	What exact turn does status buy, and what happens after miss, wake, or status block?
Physical breaker	Machop, Marowak, Belly Drum or Curse Snorlax	Converts chip/status into forced damage and punishes passive teams.	Which opposing answer must be weakened, slept, trapped, boomed, or phased first?
Recovery wall / Rest-cycle piece	Suicune, Miltank, Umbreon, Starmie, Blissey	Can stretch games, force PP/resource questions, or stall without progress if unsupported.	Is recovery preserving a route, or giving the opponent a free setup/hazard turn?

Romhack transfer rule: map the role first, then re-check the mechanics. A local species may fill a familiar role with unfamiliar stats, moves, passive type effects, items, or hazard economics.

Romhack Mechanics Checkpoints

Use these local facts when translating general singles ideas into this hack:

- Sleep duration matches vanilla Gen 2: sleep is rolled at application, stores

a **2..7** counter, and produces 1..6 asleep turns before the wake-up turn. Sleep Talk and Snore exist; Rest uses the stock Rest sleep counter.

- Spikes is not vanilla GSC. This hack has 3 layers:

1 layer is max HP / 8, 2 layers is max HP / 6, and 3 layers is max HP / 4. Flying-types are unaffected, Rapid Spin clears all layers, and a fourth Spikes click fails instead of adding value.

- Type passives and late-gen item changes mean type-chart and damage intuition

must be checked against local docs or the damage debugger before being used as a firm lab answer.

Local references:

- docs/agent_navigation/gen2_vs_modern_mechanics.md
- docs/mechanics_changes_from_base.md
- engine/battle/move_effects/spikes.asm
- engine/battle/core.asm
- tools/boss_ai_preference/README.md

Romhack Delta Register Seed

Do not transfer vanilla GSC conclusions through these mechanics without a local check:

Delta	Local fact	Strategic implication
Spikes	3 layers: max HP / 8, / 6, / 4; Flying-types ignore them; Rapid Spin clears all layers; extra layers fail after 3.	Hazard turns can be much more decisive than vanilla, but only if the setter survives long enough or the spinner is controlled. The third layer can be worth more than ordinary chip; a fourth click is a state error.
Type passives	Type-based effects apply to every mon of that type; they are not optional abilities.	Damage and status plans need local verification even when the type chart looks familiar.
Dark status shield	Full Dark blocks the first eligible incoming status; half Dark has a 50% block chance.	Do not treat sleep, Toxic, or paralysis into Dark targets as reliable one-turn control unless the shield is already consumed or the branch tolerates failure.
Dragon defender	Non-super-effective hits into Dragon are reduced by Imperial Scales.	Neutral chip into Dragon-types may be far weaker than normal; KO math must be checked before assuming a speed advantage is enough.
Dragon attacker	Dragon damaging moves can treat immunities as resistances; Dragon Outrage may use physical category when Attack exceeds Special Attack.	"Immune pivot" and Counter/Mirror Coat logic can change for Dragon routes.
Electric speed	Electric types get a small speed multiplier.	Close speed races involving Electric pivots need local speed evidence, not base-stat slogans.
Grass regrowth	Grass types heal between turns if not stasured and not full HP.	Poison/paralysis can be a resource denial tool, not merely chip or speed control.
Poison contact retaliation	Contact into Poison defenders can poison the attacker unless immunity/status/item rules block it.	Contact chip can carry hidden future cost; non-contact alternatives may preserve long-game resources.
Psychic mind shield	Psychic defenders have a small chance to take 0 damage while still registering a hit.	Low-probability defensive branches should be separated from decision quality; do not overfit to the null-damage outcome.
Steel recoil mitigation	Steel users reduce or remove recoil from recoil moves.	Recoil-based trade math is different for Steel attackers.

Resolved mismatch ledger:

- `mismatch_ground_flying_damage_001`: resolved as a Dragon's Majesty / fixture

note issue, not a debugger issue. `data/types/type_matchups.asm` correctly lists Ground into Flying as `NO_EFFECT`, and non-Dragon Ground users such as Onix, Donphan, and Golem return 0 damage into Flying targets. Dragon-typed attackers such as Dragonite, Steelix, Gyarados, and Kingdra can still hit type-chart immunities as resistances because Dragon's Majesty applies to damaging non-fixed moves. The Lance Aerodactyl fixture note "Earthquake cannot hit a Flying target" is therefore too broad for Dragonite.

Extracted delta rule: when the attacker has Dragon typing, re-check any line that relies on a type immunity. The base chart row may still say `NO_EFFECT`, but Dragon's Majesty can make the damaging move connect as resisted damage.

Simulation Notes

Scratch simulator: `.local/tmp/pokemon_singles_sim`, using canonical Pokemon Showdown-style Gen 2 mechanics through `@pkmn/sim`, installed outside tracked repo files. This is for learning general singles concepts and sanity checking strategic branches only. It does not simulate this romhack's custom moves, type passives, item changes, stat changes, trainer rosters, or boss AI. Use the repo's damage debugger, source-derived fixtures, and PyBoy/trace ROM runs before treating any result as romhack behavior or turning it into ROM scoring.

Important limitation: the current sleep/setup sims are controlled policy tests, not proof of optimal play. I picked both sides' policies: Victreebel followed the tested line and Snorlax mostly used steady Body Slam pressure. These runs show whether a line is robust against that pressure under RNG, but they do not model a strong opponent that switches, preserves counters, uses Rest strategically, or changes plans.

First sleep/setup test, 200 deterministic seeds, Gen 2 custom battle: Victreebel with Sleep Powder / Swords Dance / Sludge Bomb / Giga Drain into Snorlax using Body Slam pressure.

Observed outcomes, shown as Victreebel wins and losses across 200 simulator runs with different deterministic RNG seeds:

Line	Wins	Losses	Note
Sludge Bomb immediately	30	170	Direct damage was too slow.
Swords Dance raw, then attack	91	109	Setup helped but often ate the punish.
Sleep Powder, one Swords Dance, then attack	133	67	Best result; sleep bought the setup turn.
Sleep Powder, two Swords Dances, then attack	111	89	Worse than one boost; extra greed gave wake/miss/punish chances back.

Follow-up conditional test, shown as Victreebel wins and losses across 500 deterministic RNG seeds:

Policy	Wins	Losses	Note
Fixed script: Sleep Powder, Swords Dance, attack	334	166	Still strong, but it boosts even after a miss.
Conditional script: if sleep misses, re-score and retry only while healthy	393	107	Better; the status branch matters.

Interpretation: the user's lesson is right, and the boundary matters. Sleep is not the payoff. Sleep is a way to buy the turn that makes the payoff possible. After sleep lands, one boost can be correct if it changes the damage race; a second boost needs fresh evidence that the target cannot wake, punish, switch, or otherwise make the extra turn bad. If sleep misses, the plan has already changed; do not blindly spend the next turn on setup.

Long-battle drill, 40-turn canonical Gen 2 scratch simulation:

This was not a good-player-vs-good-player proof. It was a long-game review surface using bulky GSC-like teams. The value is in spotting strategic errors:

- Turns 1-2: Cloyster poisoned and Exploded on Starmie before setting Spikes.

Removing Starmie has value because Starmie spins and checks several threats, but sacrificing the only Spiker before hazards means the rest of the game no longer has passive switch pressure. In this romhack the cost is even higher because 3-layer Spikes can define switch economics.

- Turns 3-4: Steelix traded itself for Machamp. This removed a dangerous

breaker, but also spent the team's phazer/normal-resistant physical answer. Explosion is not just damage; it changes the future defensive map.

- Turns 5-8: Raikou used Rest while healthy and without Sleep Talk, then died

to Tyranitar while asleep. Rest is a resource cycle, not a heal button. If the sleeping turns cannot be protected or used, Rest can donate tempo.

- Turns 9-13: Exeggutor and Misdreavus combined Explosion/Perish pressure to

remove Tyranitar, but by then the team had spent Cloyster, Steelix, Raikou, and Exeggutor. Trading can be correct only if the remaining endgame is still playable.

- Turns 14-40: the game collapsed into a Snorlax mirror. Paralysis, Curse

count, Rest timing, Sleep Talk, and PP became the whole game. This is the warning: if early trades do not preserve a second route to win, Snorlax is forced to solve everything.

Extracted long-game rule: every sacrifice must name the future it buys. "I removed Starmie" is incomplete unless the next sentence is "therefore my hazard, Machamp, Marowak, Raikou, or Snorlax route is now live." If the trade only makes the current turn look good while deleting the support for the endgame, it is probably bad.

Second long-battle drill, 21-turn canonical Gen 2 scratch simulation:

Team A used Cloyster / Raikou / Snorlax / Steelix / Exeggutor / Starmie into Forretress / Zapdos / Snorlax / Skarmory / Machamp / Misdreavus. The script uses seeded RandomPlayerAI policies, so it is reproducible, but still only a mistake-finding surface rather than expert proof.

Critical findings:

- Turns 1-3: both sides committed early to Spikes plus Explosion. Forretress

set Spikes and blew up; Cloyster set Spikes and blew up on Machop. This was not automatically bad because Team A's Spikes immediately taxed Machop, Snorlax, and Misdreavus. The question is whether the trade names its future: "I am spending the Spiker because the hazard is already down and removing Machop opens the Snorlax/Raikou route."

- Turn 4 and turn 20: Zapdos and Raikou clicked Rest while Rest could not

produce useful recovery. This is a state-precondition error. Recovery has to preserve a route or change a damage threshold; otherwise it is just a donated turn.

- Turns 5-9: Steelix used Toxic plus Explosion to put Snorlax into a forced

collapse sequence. Spikes, poison, Explosion chip, Rapid Spin tempo, Starmie damage, and Exeggutor damage together removed the central threat. This is a good example of a cash-out line: no single move solved Snorlax, but each move converted the previous resource into the next one.

- Turn 10: Exeggutor Exploded into Misdreavus Protect and got nothing. This is

the opposite of the Steelix line. Explosion is only a cash-out when the target is forced, trapped, sufficiently punished for switching, or the free entry is itself the named payoff.

- Turns 11-16: Snorlax was trapped by Mean Look and then lost to the Perish

Song clock. Once the trap route starts, normal Rest/Curse logic is no longer the main game. The position becomes a clock problem: switch before the trap if possible, KO or phaze before the count expires, or accept the trade only if the remaining endgame is winning.

Extracted long-game rule: a status move is only progress if it changes the next resource exchange. Sleep, poison, and hazards all need a follow-up route. If the next turns do not cash the created opening, the opponent can convert the "control" turn back into Rest, setup, or a favorable trade.

Third long-battle drill, 81-turn canonical Gen 2 scratch simulation:

Batch run: `node .local/tmp/pokemon_singles_sim/long_gsc_drill.js batch 60`. Seed offset 36 reached 81 turns and ended with Ledger B winning.

Critical findings:

- Turns 1-3: Cloyster clicked Toxic twice into Forretress, then Exploded into a

Steel-type that survived. This is two mistakes chained together: status into an immune target, then Explosion without a forced or sufficiently valuable target. Cloyster died without setting Spikes, so Team A lost its Spiker and got no hazard economy in return.

- Turns 5-6: Raikou used Sleep Talk while awake, then stayed in and died to

Snorlax Earthquake after Thunder missed. The miss was bad luck; staying in with no protection against the obvious Ground punish was the decision issue.

- Turns 11-12: Starmie used Rapid Spin when there were no Spikes to remove,

then died to Machamp. Rapid Spin can be valid chip only if the chip matters; otherwise it is another state-precondition failure.

- Turns 25-77: Snorlax eventually beat Skarmory, but it took more than 50

turns of Curse, Toxic, Rest, Whirlwind, and Body Slam pressure. This was not clean progress; it was a single-route endgame grinding through a dedicated phazer.

- Turn 49: Exeggutor Exploded into sleeping Skarmory for partial damage. That

helped Snorlax eventually remove Skarmory, but it also deleted Team A's last non-Snorlax piece.

- Turns 78-81: preserved Misdreavus used Perish Song to force the final trade

and win. The final error was prepared earlier: Team A reduced itself to one Snorlax route before checking whether the opponent still had a Ghost/Perish endgame.

Extracted long-game rule: before spending the second-to-last route, name the opponent's preserved anti-route. A sacrifice that helps win the current wall war can still lose the match if it leaves one Pokemon against an unrevealed or preserved endgame answer.

Fourth long-battle drill, filtered-policy seed 15, 76-turn canonical Gen 2 scratch simulation:

This game used the filtered random policy, so the obvious state-invalid moves were already removed. The remaining lessons are higher order:

- Turns 1-5: Cloyster set Spikes and then Exploded into Forretress, but

Forretress survived, spun the Spikes away, and then Exploded on Raikou. This is a sharper version of the sacrifice rule: even after a hazard is placed, spending the Spiker is bad if the opponent's hazard-control piece survives and removes the hazard anyway.

- Turns 9-13: Misdreavus trapped Starmie and used Perish Song. Starmie removed

Misdreavus before the clock expired, but still fainted to the Perish count. This was a trade, not a clean win. When a trap clock starts, the question is not just "can I KO the trapper?" but "what does the forced trade leave me against?"

- Turns 17-20: Exeggutor landed Sleep Powder on Skarmory, then used Giga Drain

twice before Explosion. This is the corrected version of the earlier sleep mistake: once sleep lands, use the created turns for damage, switch, setup, or a forced trade; do not keep clicking sleep into a sleeping target.

- Turns 21-76: the game became a Snorlax mirror into Zapdos and Machamp. Team A

eventually won because the opponent's Misdreavus was gone, so collapsing to a Snorlax route was not punished the way it was in drill 3. This is the important comparison: one-route endgames are bad only when the opponent still has the anti-route.

Extracted long-game rule: a sacrifice can be correct in one battle and losing in another with the same Pokemon if the preserved anti-route differs. The ledger must ask "what remains after this trade?" before judging the trade.

Fifth long-battle drill, canonical Gen 2 scratch simulation, seed 0, filtered policy:

- The hard state gates were clean: no awake Sleep Talk, no status into public

immunity/status, no Rapid Spin with no hazards, no full-HP Rest, and no Explosion into a known Ghost.

- Turns 1-4: Cloyster set Spikes, then Exploded into Forretress. This time the

trade was converted because Raikou immediately removed Forretress before it could spin. The same Spiker cashout that was bad in drill 4 became playable because hazard control did not survive.

- Turns 6-34: Exeggutor traded Explosion for Machop, then Snorlax used

Body Slam / Curse / RestTalk to remove Zapdos and win the Snorlax mirror. That is a real route, but it also reduced the game toward one main route.

- Turns 36-52: Skarmory phazed Snorlax repeatedly while Spikes punished the

dragged bench. Starmie eventually used Rapid Spin with a real hazard-control job, clearing Ledger A's side before dying later.

- Turns 64-74: Snorlax converted through Skarmory only after forcing repeated

Rest cycles. This was not a simple "boost and attack" line; it depended on Rest timing, paralysis, Whirlwind turns, and keeping enough HP to survive until the phazer was gone.

- Turns 76-79: Misdreavus entered last and used Perish Song / Protect /

Mean Look. The simulator ended with both final Pokemon fainting and no winner line. Strategically, the lesson is still clear: a last-route Snorlax that cannot hit Ghosts cannot treat a preserved Misdreavus as a normal damage endgame.

Extracted long-game rule: a route is not complete just because it beats the current wall. Before committing to a last-Pokemon Snorlax line, check whether the opponent's preserved Ghost, trap clock, phazer, or Explosion route can still invalidate it after the apparent wall is removed.

Local Rest source check:

- `engine/battle/effect_commands.asm` compares current HP to max HP and jumps

to the `.hp_full` failed-move path before the Rest-specific status cure and sleep-counter code.

- Therefore, in the local engine, Rest at exactly full HP fails even when the

user is statused. Status-cure Rest requires the user to be below full HP, and the resulting sleep turns still need to be playable.

Measured state-error regression, same scratch simulator:

Command:

```
node .local/tmp/pokemon_singles_sim/long_gsc_drill.js analyze 200 random
node .local/tmp/pokemon_singles_sim/long_gsc_drill.js analyze 200 filtered
```

The **filtered** policy is still random after applying only hard precondition gates: no awake Sleep Talk, no direct Rest at exactly full HP, no direct Rest at high HP with no status, no Rapid Spin with no own Spikes, no Toxic or sleep move into a target that was immune or already statused at turn start, and no Explosion into a known Ghost / Misdreavus denial target. It is not a strong player. It is a test of whether the notebook's hard gates remove obvious state errors.

Metric across 200 games	Random policy	Filtered policy	Interpretation
Total turns	8292	6612	Different games, so compare error classes, not win rate.
Awake Sleep Talk	668	0	Hard state gate works.
Rapid Spin with no hazards to remove	412	0	Hazard-state gate works.
Toxic into immune/statused target	360	0	Status target gate works.
Sleep move into already-statused target	64	0	Sleep gate works when judged by turn-start state.
Direct Rest at exactly full HP	798	0	Local source-backed hard gate works.
Direct Rest at exactly full HP while statused	350	0	Status does not rescue full-HP Rest because the full-HP failure path comes first.
Direct Rest at high/full HP with no status	545	0	Hard recovery gate works.
Direct Rest at high/full HP while statused	501	33	Remaining cases are below full HP; they need Rest-cycle review, not a hard ban.
Explosion denied without target damage	52	0	Ghost / known denial target gate works; broader Explosion timing still needs judgment.

Extracted measurable target: the next policy layer should keep the hard state metrics at zero while learning when below-full status-cure Rest, hazard sacrifice, and one-route endgames are actually worth the tempo.

Review of the remaining below-full, high-HP status Rest cases:

- Seed 58 turn 11: Snorlax used Rest at 432/460 while toxic, facing a sleeping

Skarmory. This is a plausible status-cure Rest because the move succeeds, removes a worsening poison clock, and Sleep Talk gives the sleeping turns a job.

- Seed 1 turn 12: Raikou used Rest at 276/320 while toxic, but opposing

Snorlax immediately used Double-Edge and KOed it. This is not a Rest-cycle success; the status cure never matters if the public damage branch removes the user first.

- Seed 16 turn 16: Snorlax used Rest at 431/460 while toxic, then Skarmory

Whirlwinded it into a Spikes-taxed Raikou. The Rest clears status, but the phaze branch changes the resource exchange. Status-cure Rest must check phazing and forced-entry consequences, not only HP and status.

Extracted Rest rule: below-full status-cure Rest is allowed, but only after checking three branches: immediate damage before recovery, phazing/trapping after recovery, and whether the sleep turns have Sleep Talk or safe board support.

Follow-up filtered-policy regression:

```
node .local/tmp/pokemon_singles_sim/long_gsc_drill.js analyze 300 filtered
```

Across 300 filtered games, the hard-invalid counters stayed at zero: awake Sleep Talk, Rapid Spin with no hazards, Toxic into immune/statused targets, sleep into already-statused targets, full-HP Rest, high-HP no-status Rest, and Explosion into known Ghost denial. The only remaining flagged class was 53 below-full, high-HP status Rest events. That supports the split now used in this notebook: full-HP Rest is a mechanics/state gate; below-full status-cure Rest is a tactical judgment that must be reviewed through immediate damage, phazing/trapping, and playable sleep turns.

Large-sample correction:

```
node .local/tmp/pokemon_singles_sim/long_gsc_drill.js analyze 1000 random
node .local/tmp/pokemon_singles_sim/long_gsc_drill.js analyze 1000 filtered
```

The 1000-game random baseline produced 3320 awake Sleep Talk attempts, 2046 Rapid Spins with no hazards, 1818 Toxic attempts into immune/statused targets, 3839 full-HP Rest attempts, and 238 Explosion denial events. The 1000-game filtered policy still held most hard-invalid classes at zero, but exposed 4 full-HP Rest attempts in very long games and 177 below-full high-HP status Rest events. Manual inspection of seeds 759 and 577 showed these rare full-HP Rest cases around long RestTalk, phazing, trap-clock, PP, or failed-move endgames.

Correction to the target: do not claim full-HP Rest is completely solved by the current scratch policy. The rule is still mechanically true, but the simulator policy needs a PP-aware fallback and failed-move context so it does not choose a known failed recovery move when every attractive line has collapsed.

Follow-up simulator parser correction:

```
node .local/tmp/pokemon_singles_sim/long_gsc_drill.js analyze 500 random
node .local/tmp/pokemon_singles_sim/long_gsc_drill.js analyze 500 filtered
```

The scratch ledger was over-counting some sleep errors because it kept old status when a later HP condition had no status suffix. The parser now treats statusless HP conditions as public status clears and treats successful non-Sleep Talk moves as wake evidence. This matters because stale `slp` state can make the policy think Sleep Talk or sleep immunity is legal when the Pokemon has already woken.

After the parser/status correction, the 500-game comparison was:

Metric across 500 games	Random policy	Filtered policy	Interpretation
Total turns	20443	16468	Different games; compare error classes only.
Awake Sleep Talk	2447	1	Hard gate is almost complete; one residual long-game fallback remains to audit.
Rapid Spin with no hazards to remove	1018	0	Hazard-state gate held.
Toxic into immune/statused target	780	0	Status target gate held.
Sleep move into already-statused target	123	0	Sleep target gate held after wake-state cleanup.
Direct Rest at exactly full HP	1942	2	Still not fully solved in rare fallback states.
Direct Rest at high/full HP with no status	2387	2	No-status Rest is nearly eliminated but not proven gone.
Direct Rest at high/full HP while statused	222	63	These need branch review, not a blanket ban.
High/full HP Rest while toxic	171	39	Often a plausible cure line if the sleep turns are playable.
High/full HP Rest while paralyzed	44	15	More suspect than Toxic; must justify speed/status cure value.
Manual Rest while already asleep	0	0	The previous nonzero count was stale ledger state, not confirmed play.
Explosion denied without target damage	129	0	Known Ghost denial gate held.

Extracted simulator rule: state cleanup is a first-class mechanic. When a Showdown HP condition loses its status suffix or a sleeping Pokemon successfully uses a non-Sleep Talk move, clear the old status in the ledger. Then evaluate Rest by status class: full-HP Rest is a mechanics failure, high-HP no-status Rest is usually wasted tempo, Toxic-cure Rest can be correct, paralysis-cure Rest needs a stronger speed or endgame reason, and every below-full Rest must still price immediate damage, phazing/trapping, and playable sleep turns.

Follow-up implementation check: splitting the scratch policy into hard-invalid and soft-avoid move classes, then reading the active slot more carefully, did not change the 1000-game counts. Inspecting seed 759's ending explains why. Snorlax was trapped by Misdirection, Body Slam could not affect the Ghost, sleep ended at full HP, the Perish count was already running, and the remaining move choices were failed or exhausted endgame actions. Future reports must separate avoidable full-HP Rest from no-exit failed-move endgames where the real mistake was earlier: preserving no answer to the trap clock.

Sixth long-battle drill, filtered-policy seed 264, 79-turn canonical Gen 2 scratch simulation:

- Turns 1-7: Cloyster set Spikes and Exploded into Forretress. Starmie then

spun away Forretress's Spikes and was removed by Forretress Explosion. This was a trade chain, not a simple hazard win: both sides spent hazard-control pieces, and the resulting game no longer had easy long-term Spikes pressure.

- Turns 8-29: Steelix and Exeggutor were spent into Skarmory, then Snorlax and

Raikou finally removed Skarmory. The route worked only because Raikou could cash the phazer after Snorlax forced damage and Rest cycles. The review question is whether each earlier Explosion made that route easier or merely narrowed the game to one route.

- Turns 30-52: Raikou versus Zapdos became a RestTalk and PP/damage-range

cycle. Rest was playable when the sleeping turns either had Sleep Talk or the opponent was also forced into recovery. Rest is bad when it donates tempo; it is acceptable when it preserves the only Electric route through a mirrored recovery cycle.

- Turns 60-64: low-HP Raikou was caught by Misdreavus's Perish Song / Mean

Look line and traded down. This was acceptable only because Snorlax still had a real endgame into the opposing Snorlax afterward. If the remaining endgame were worse, the earlier choice to leave Misdreavus preserved would have been the irreversible error.

- Turns 65-79: Snorlax won the final mirror by combining Body Slam paralysis,

Curse timing, and attacking through Rest turns. The game was decided by whether the final route survived all earlier trades, not by one flashy move.

Extracted seed-264 rule: before accepting a chain of sacrifice trades, name the final route left after the chain. "I eventually win with Snorlax" is only valid if the remaining opposing Ghost, phazer, Explosion, RestTalk, and PP routes have been checked.

First external replay review, 46-turn GSC OU game:

- Source context: Smogon GSC Gyarados discussion, proof replay linked from a

July 2025 post: <https://www.smogon.com/forums/threads/gsc-ou-discussion-thread.3688141/> and replay log <https://replay.pokemonshowdown.com/gen2ou-2404705604-gl00guvgmu8l6etle1dd06927fn0jlsplpw.log>.

- Opening, turns 1-4: Zapdos and Snorlax traded heavy damage, then both used

Rest. The immediate damage race reset, but the sleep turns created a new positional phase instead of ending the exchange.

- First conversion attempt, turns 8-10: JackRG1 used Belly Drum Snorlax, but

jouer no L preserved the correct answer by going Skarmory. The boosted Return did not break Skarmory, and Snorlax had to leave. Setup failed because the opponent's defensive answer was still available.

- Hazard phase, turns 11-14 and 21-23: Cloyster established Spikes, but the

game still required hazard control. Forretress later set Spikes back against a sleeping Zapdos, showing how Rest can donate a hazard turn if the sleeping turns are not actively useful.

- Gyarados phase, turns 15-19 and 29-32: Gyarados used Surf and Toxic to force

awkward pivots and chip Snorlax/Machop, but it was not a solo breaker. Its value was role compression and forcing sequencing, not sweeping.

- Spin phase, turns 32 and 38: Golem removed one side's Spikes, then Forretress

removed the other side's Spikes. A Spikes plan is not complete when the layer goes down; it is complete only if the player can keep the layer relevant or make the spin costly.

- Final conversion attempt, turns 39-46: JackRG1 tried Belly Drum Snorlax

again after forcing Skarmory down. The line still failed because the opposing Snorlax was asleep on the bench, returned near full, and woke/acted in time to KO the Drummed Snorlax with Double-Edge.

Extracted external-review rule: sleep state is not a permanent disable and bench sleep can still return as a wake-turn punish. Before setting up into a sleeping or recently sleeping target, track whether the target can wake, switch back in, phaze, trade, or attack before the setup route converts.

Replay review helper:

- Scratch tool: `.local/tmp/pokemon_singles_sim/replay_ledger.js`.
- Example:

```
node .local/tmp/pokemon_singles_sim/replay_ledger.js <showdown-log-url>.
```

- Purpose: turn Showdown logs into a critical-turn ledger tagged for setup,

Rest/recovery, Sleep Talk, hazards, spin, phazing, sacrifice, trap clocks, sleep punishment, switches, and faints. This does not judge the best move by itself; it makes long-game review less likely to skip important resource turns.

Second external replay review, 71-turn GSC OU game:

- Source context: Smogon GSC OU discussion replay list:

<https://www.smogon.com/forums/threads/gsc-ou-discussion-thread.3688141/> and replay log
<https://replay.pokemonshowdown.com/smogtours-gen2ou-505883.log>.

- Turns 1-16: early Curse Snorlax forced Skarmory phazing, then both players

established Spikes and traded hazard control. Golem spun, Cloyster Exploded, and Snorlax was removed. This is a clear route-trade sequence: hazard control and Explosion were used to change which late-game pieces mattered.

- Turns 19-30: Raikou Rested, Forretress used the sleeping/electric pressure

window to re-enter the hazard game, Gengar landed sleep pressure into Zapdos, then Exploded into Raikou. The lesson is not simply "Explosion good"; it removed the RestTalk Electric and left the game to Snorlax/Skarmory/Golem structures.

- Turns 32-58: Skarmory, Snorlax, and Golem entered a long phazing and Rest

cycle. Golem's Roar repeatedly denied Skarmory's Curse plan while Earthquake punished the forced Snorlax entries. This is the kind of 20+ turn sequence the notebook must learn: the good move is often the one that keeps the loop favorable, not the one that deals the most immediate damage.

- Turn 59: simultaneous phazing mattered. Golem's Roar failed while Skarmory's

Whirlwind dragged Starmie in, creating the route that later ended the game.

- Turns 68-71: Golem Exploded into Skarmory, Starmie cleaned up with Surf, and

Nightmare punished the sleeping Snorlax before the final KO.

Extracted external-review rule: when a long game becomes a phazing loop, track which side benefits from every forced entry. The loop is progress only if it changes HP, sleep turns, hazard state, PP, or final-entry access in favor of a named endgame route.

Third external replay review, 54-turn GSC OU game:

- Source context: Smogon GSC OU Winter Seasonal #8 Round 4, leoperi vs

DonMarto result post: <https://www.smogon.com/forums/threads/gsc-ou-winter-seasonal-8-round-4.3778203/> and replay log <https://replay.pokemonshowdown.com/smogtours-gen2ou-912130.log>.

- Turns 1-5: don marto opened with Curse Snorlax, but leoperi immediately used

Lovely Kiss, then both sides established Spikes. The early Curse did not become a sweep route because the Snorlax route was paused by sleep and then taxed by hazards for the rest of the game.

- Turns 6-18: Spikes converted ordinary pivots into real resource loss.

Meganium repeatedly absorbed Hidden Power / Earthquake pressure and used Synthesis and Light Screen to keep the defensive route intact. Chip was only progress when it forced a non-recoverable state; otherwise Meganium turned it back into a tempo question.

- Turns 24-28: leoperi's Zapdos Rested, then Sleep Talk called Rest at 27% and

reset the damage race to full. The important lesson is mechanical and strategic: attacking a sleeping RestTalk user is not a guaranteed conversion unless the Sleep Talk Rest branch has been priced in.

- Turns 33-41: Golem spun away don marto's Spikes, but leoperi preserved enough

hazard access to re-layer with a poisoned Cloyster on turn 41. Rapid Spin was a temporary resource win, not permanent control, because the opposing Spiker still had a final job.

- Turns 42-43: leoperi's Cloyster completed a compact route trade. It set

Spikes, Exploded into opposing Cloyster, then Houndoom cleaned that Cloyster. This was a sacrifice with two concrete jobs: restore switch tax and remove hazard control / Water counterplay.

- Turns 43-54: the early-slept Snorlax finally returned, still asleep, took

Spikes and Leech Seed pressure, and woke only in time to Earthquake Rhydon before dying to boosted Earthquake. Rhydon then removed sleeping Zapdos before falling to Moltres, leaving leoperi's Zapdos to finish. The endgame was won because leoperi's final Zapdos route survived the Rhydon trade.

Extracted external-review rule: a preserved sleeping win condition is not the same thing as an available win condition. Track how many turns it needs before acting, what entry damage it must absorb, whether the opponent can set up while it sleeps, and whether the wake turn actually arrives before the route is lost.

External source synthesis, GSC role compression:

- Source context:
 - Cloyster analysis thread:

<https://www.smogon.com/forums/threads/gsc-ou-cloyster.3708121/>

- Snorlax analysis thread:

<https://www.smogon.com/forums/threads/gsc-ou-snorlax.3694359/>

- Steelix analysis thread:

<https://www.smogon.com/forums/threads/gsc-ou-steelix.3705042/>

- Cloyster is not just "the Spikes Pokemon." The source frames Spikes as a way

to make repeated grounded answers like Snorlax and Raikou stop entering for free, while Surf/Toxic/Explosion make spin attempts and Explosion pivots costly. Heuristic implication: after setting Spikes, ask what makes the opponent's removal or grounded pivot line expensive.

- Steelix turns the hazard layer into a repeated decision point through Roar.

The source explicitly ties Roar, Spikes damage, and Cloyster partnership together. Heuristic implication: a phazing turn should be scored by the forced entry it creates, not by whether it looks passive.

- Snorlax's Curse + Double-Edge/Earthquake pressure can force phazing or win

outright, but Rest without Sleep Talk creates a setup/phazing window and makes hazard control more important. Heuristic implication: Rest is not just healing; it changes who gets to spend the next two turns.

- Combined rule: in long GSC games, role compression matters more than species

slogans. A Cloyster turn, Steelix turn, or Snorlax Rest turn must be judged by how it changes hazard relevance, phazing access, Explosion routes, and the remaining endgame pieces.

Current Learning Rules

- Think in routes to win, not isolated best moves. A move is good when it keeps or improves the route.
- Prefer the line that wins against normal play and remains acceptable against the worst plausible punish.
- A KO is not automatically best if it gives the opponent a free setup entry that wins next turn.
- Setup needs a reason: it must change KO math, force a defensive response, or make a later sweep possible.
- Sleep is turn creation. Use it when the created turn has a clear job, such as one boost, one safe switch, or one recovery denial.
- Do not autopilot after a status move. If sleep misses, if the target wakes, or if Sleep Clause blocks another sleep, re-score the board.
- A preservation switch is good only when the saved Pokemon has real future value and the switch-in is actually safer.
- Long games are resource games. HP, PP, status slots, revealed moves, hazards, and future switch paths all matter.
- Reject state-invalid moves before taste scoring. Awake Sleep Talk, blocked sleep, repeated status into an already-statused target, and setup after the setup window has closed are not real plans.
- Treat full-HP Rest as state-invalid in the local engine. Rest can cure status only if the user is below full HP, and it is still bad if the sleep turns give away the position.

Turn Ledger Template

Use this before reviewing or simulating a serious turn. If I cannot fill this out, I do not know the position well enough to call the move "optimal".


```
Battle / fixture:
Turn:
Format evidence: GSC / romhack source / romhack runtime / unknown

Own active:
Opponent active:
Own bench still relevant:
Opponent bench still relevant:

HP ranges:
Status:
Sleep or Rest counters:
Hazards:
Screens / weather / volatile state:
Stat stages:
Known moves:
Likely unrevealed moves:
PP pressure:
Speed order:
Damage ranges that matter:

Own current win routes:
Opponent current win routes:
Own irreplaceable pieces:
Opponent irreplaceable pieces:

Candidate 1:
Immediate result:
Opponent stay-in punish:
Opponent switch punish:
Opponent setup / Rest / status / Explosion punish:
Low-roll / miss / wake / full-para branch:
Resource gained:
Resource spent:
Next-turn plan:
Verdict:

Candidate 2:
Candidate 3:
```

The ledger should force concrete language. "Momentum" is not a resource unless it names the follow-up: a safe Rest turn, a forced switch, a Spikes layer, a phaze loop, a damage range, a preserved answer, or a revealed set.

Decision Classifier

After ranking candidates, label the chosen move with one primary class:

Class	Use when	Failure sign
Force	The move wins now, prevents an immediate loss, or creates a forced sequence.	I ignore a reliable force to chase style points.
Convert	The move turns an existing edge into a clearer route: KO range, forced Rest, hazard loop, or endgame simplification.	I take chip that does not change any later turn.
Preserve	The move keeps an irreplaceable answer alive or avoids exposing a losing route.	I save a Pokemon without a named future job.
Scout	The move gathers information while keeping the dangerous branches covered.	I scout when the opponent already has a forced punish.
Deny	The move blocks setup, spin, recovery, sleep, phazing, or a free switch.	I deny something that was not actually threatening.
Cash out	The move spends a resource, including Explosion or sacrifice, to open a named route.	The trade removes my own last answer or leaves no route.

If no class fits, the move is probably a vague "do something" click and needs to be re-scored.

Heuristic Card Template

Promote a rule only when it can choose moves in positions like the lab asks.

Heuristic: [short name]

Status: proposed / developing / stable / rejected

Format: GSC / romhack / both / unverified transfer

Evidence tier: source / fixture / runtime / simulation / user answer

Last reviewed:

Trigger:

- Exact board features that make the rule relevant.

Rule:

- The move preference or decision policy.

Why this should be true:

- Mechanics, damage ranges, role map, or long-term resource reason.

Candidate lines:

- Best line:
- Main alternative:
- Emergency line:

Exceptions:

- Conditions that flip the decision.

Worst plausible punish:

- The branch that would make this line bad.

Evidence:

- Source:
- Fixture:
- Damage check:
- Simulation:
- Battle review:

Counterexamples:

- Positions where the rule fails or becomes lower priority.

Benchmark tests:

- Scenario IDs that must keep passing.

Failure signs:

- GPT-shaped phrases, missing board facts, or known repeated mistakes.

Current stable example:

```
# Heuristic: Re-score after sleep disruption

Status: stable
Format: both, with romhack legality checks
Evidence tier: user answer + simulation + fixture damage

Trigger:
- A sleep move misses, the target wakes, Sleep Clause blocks sleep, the target switches, Sleep Talk reveals a dangerous move, or new damage information changes the matchup.

Rule:
- Discard the prior sleep/setup script and recompute from the current board.
- Setup is still legal only if it changes KO math or creates a forced route after checking the new worst plausible punish.

Exceptions:
- Continue the prior plan only when the failed or changed branch still leaves no immediate KO, setup, phazing, or defensive-answer-loss punish.

Failure signs:
- Saying "continue the plan" without naming the changed state.
- Boosting after a miss because the original line wanted a boost.
```

Battle Review Protocol

For 30+ turn games, review in phases instead of trying to explain every click equally.

1. Reconstruct the ledger for every turn that changes resources: HP ranges, status, sleep or Rest counters, hazards, PP, revealed moves, boosts, and role obligations.
1. Split the game into opening, midgame, conversion, and endgame. The boundary is not the turn number; it is when the win routes or resource priorities change.
1. Mark critical turns where the player could set hazards, spin, sleep, wake, Rest, phaze, setup, sacrifice, explode, preserve an answer, or reveal coverage.
1. For each critical turn, compare at least three candidate lines with the turn ledger template.

1. Separate outcome from decision quality:

- good decision / bad outcome;
- bad decision / good outcome;
- mechanics error;
- state-tracking error;
- damage-range error;
- win-condition error;
- sacrifice error;
- script-following error;
- romhack-transfer error.

2. Find the earliest irreversible error, not just the final losing move.**3.** Extract exactly one or more notebook updates: a promoted heuristic, a

narrowed exception, a rejected rule, or a new benchmark scenario.

Mistake Loop

Every serious mistake should leave a regression trail:

```
mistake observed
-> exact position captured
-> error class assigned
-> hypothesis written
-> controlled test or fixture chosen
-> result recorded
-> heuristic changed
-> benchmark scenario added
-> future reviews checked for recurrence
```

Acceptable mistake note:

- "After Sleep Powder missed, I kept the Swords Dance script and ignored that

Body Slam now put Victreebel into range. Add `sleep_disruption_001` and require re-scoring after miss/wake/switch."

Unacceptable mistake note:

- "Be more careful with sleep."

Mistake Ledger Seed

These are the first recurring error patterns to regress against.

ID	Error class	Mistake pattern	Current regression
mistake_sleep_script_001	script-following error	After a sleep miss, wake, switch, or Sleep Clause block, continuing the old setup line as if sleep succeeded.	bench_sleep_setup_erika_001
mistake_spiker_cashout_001	sacrifice error	Spending the only Spiker before hazards are established without naming the future route bought by the trade.	Hazard war drills; long-battle drill turns 1-2
mistake_rest_tempo_001	tempo error	Resting while healthy or unprotected, then donating sleep turns to a breaker or setup threat.	30+ turn battle reviews; Rest timing drills
mistake_rest_full_hp_status_001	mechanics/state error	Treating full-HP Rest as a status cure even though the local engine fails the move before clearing status.	Local Rest source check; filtered-policy state gat
mistake_setup_axis_001	setup tempo error	Choosing a defensive setup move because it boosts the right stat, while ignoring that the user is slower and already in KO range.	bench_setup_tempo_will_001
mistake_hazard_greed_001	hazard tempo error	Clicking Spikes because hazards are valuable while ignoring a faster revealed KO on the setter.	bench_hazard_tempo_koga_001
mistake_sacrifice_no_future_001	sacrifice error	Exploding or sacrificing because the current turn improves,	bench_cashout_brock_001

ID	Error class	Mistake pattern	Current regression
		without naming the next entry and the route opened.	
mistake_species_slogan_001	role-map error	Saying "this Pokemon walls/checks/sets hazards" without checking HP, speed, revealed moves, hazards, and whether that role is still needed.	Turn ledger and role map checks
mistake_state_invalid_001	state-tracking error	Considering a move whose current-state precondition is false, such as awake Sleep Talk or repeated status into an already-statused target.	Long-battle drill 2; state-invalid move checks
mistake_status_no_payoff_001	conversion error	Landing sleep, poison, or hazards without a named follow-up that converts the created turn or passive damage into progress.	Long-battle drill 2; sleep/status drills
mistake_boom_unforced_001	sacrifice error	Exploding into a target that can Protect, switch, resist, or ignore it without having made that branch costly.	Long-battle drill 2 turn 10; Explosion drills
mistake_trap_clock_001	endgame clock error	Treating Mean Look plus Perish Song as a normal damage race instead of a forced countdown that changes every priority.	Long-battle drill 2 turns 11-16; endgame drills

ID	Error class	Mistake pattern	Current regression
mistake_no_exit_endgame_001	win-condition error	Reaching a final state where all remaining choices are failed moves, immunities, or exhausted PP because the earlier route map did not preserve an exit from trap/Perish/Ghost endgames.	Filtered-policy seed 759 turns 93-108; endgame drills
mistake_sleep_wake_setup_001	setup / state-tracking error	Treating a sleeping bench target as permanently disabled and setting up into a wake-turn punish.	bench_external_belly_drum_wake_00
mistake_phaze_loop_progress_001	long-game planning error	Reviewing a phazing loop as repeated noise instead of tracking which side gains HP, sleep turns, PP, hazard control, and final-entry access.	bench_external_phaze_loop_golem_s
mistake_status_immune_001	mechanics error	Clicking Toxic or another status into a target publicly immune to that status.	Long-battle drill 3 turns 1-2; mechanics oracle
mistake_sleep_statused_001	state-tracking error	Clicking a sleep move into a target that was already statused at turn start.	Filtered-policy state gate; sleep/status drills
mistake_spin_no_hazard_001	state-tracking error	Clicking Rapid Spin as if hazard control is needed when there are no hazards to remove and the chip does not matter.	Long-battle drill 3 turn 11; hazard-control drills

ID	Error class	Mistake pattern	Current regression
mistake_single_route_endgame_001	win-condition error	Spending the second-to-last route before checking whether the opponent's preserved final piece beats the only remaining route.	Long-battle drill 3 turns 49 and 78-81
mistake_hazard_control_survives_001	conversion error	Spending the Spiker after setting hazards while the opponent's spinner survives and can remove them before the hazard pressure matters.	Filtered-policy drill 4 turns 1-5
mistake_restattalk_rest_branch_001	mechanics / conversion error	Treating attacks into a sleeping RestTalk user as guaranteed progress without pricing the Sleep Talk -> Rest branch that can reset the damage race.	bench_external_restattalk_zapdos_00
mistake_spin_temporary_control_001	hazard-control error	Treating Rapid Spin as final hazard control while the opposing Spiker is still alive or can trade itself to re-layer.	bench_external_spin_relayer_cloys
mistake_sleeping_wincon_available_001	win-condition error	Counting a sleeping preserved Pokemon as an active endgame route without tracking entry damage, sleep turns, setup pressure, and whether the wake turn arrives in time	bench_external_sleeping_lax_endga

Benchmark Suite Backlog

Build these sets before claiming mastery. Each benchmark must have a held-out side so the notebook does not merely memorize its own examples.

Suite	Purpose	First evidence source
Mechanics oracle	Verify sleep, Rest, Sleep Talk, phazing, Explosion, Spikes, speed, and romhack deltas.	Local mechanics docs and source.
Curated one-turn positions	Check whether the decision checklist picks the right move class.	Preference fixtures.
Three-turn branch positions	Test sleep/setup, switch/preserve, Rest timing, and punish branches.	Fixture variants plus damage debugger.
30+ turn battle reviews	Practice long-game ledgers, turning-point analysis, and endgame planning.	Showdown GSC logs or scratch sims.
Hazard war drills	Compare no Spikes, vanilla one-layer Spikes, and romhack three-layer Spikes.	Local Spikes docs and custom fixtures.
Sacrifice / Explosion drills	Require naming what future a trade buys and what answer it deletes.	GSC Explosion guide plus fixture variants.
Snorlax route drills	Preserve multiple answers and avoid assuming one check covers all sets.	GSC threatlist plus local trainer data.
Romhack delta drills	Re-test any vanilla conclusion touched by type passives, items, hazards, learnsets, or AI constraints.	<code>docs/agent_navigation</code> and runtime checks.

Initial pass gate:

- no critical mechanics mistake in the scenario explanation;
- best move class within the accepted answer set;
- at least two alternatives considered;
- worst plausible punish named;
- current win route and opponent win route named;
- no unverified Showdown result used as romhack proof.

State-Transition Benchmark Policy

The active goal direction is now tested state-transition policy, not broad notebook prose. Heuristics should be promoted only when they can survive a hard position that forces move choice, arbitration, hidden-info reasoning, and catastrophe analysis.

Machine-readable benchmarks are split so the policy cannot see the answer key:

- public cards:

`tools/boss_ai_preference/benchmarks/state_transition_public_cards.json`

- hidden oracles:

```
tools/boss_ai_preference/benchmarks/state_transition_oracles.json
```

- legacy combined source artifact:

```
tools/boss_ai_preference/benchmarks/state_transition_benchmarks.json
```

- policy answer schema:

```
tools/boss_ai_preference/schemas/policy_answer.schema.json
```

`tools/boss_ai_preference/benchmark_positions.py` validates the public cards, hidden oracles, and structured policy answers. The deterministic baseline consumes only public cards. The evaluator joins hidden oracle labels by benchmark id when scoring the answer file. Structured policy answers must carry non-empty route, preservation, catastrophe, answer-changing information, and fired-rule fields. They must not carry hidden oracle labels such as `best`, `acceptable`, `catastrophic`, or oracle rationale. Run `python -m tools.boss_ai_preference benchmark-policy` to write the current deterministic baseline policy answers to `audit/boss_ai_preference/state_transition_policy_answers.json`, then run `python -m tools.boss_ai_preference benchmark-report --answers audit/boss_ai_preference/state_transition_policy_answers.json` to write the current benchmark contract and policy-answer evaluation to `audit/boss_ai_preference/state_transition_benchmarks.md` and `audit/boss_ai_preference/state_transition_benchmarks.json`. Run `python -m tools.boss_ai_preference benchmark-mutations` to generate the current boundary-mutation audit at `audit/boss_ai_preference/state_transition_mutations.md` and `audit/boss_ai_preference/state_transition_mutations.json`. These mutations are not policy inputs; they are evaluator checks that a minimal public-state delta flips the deterministic answer for sleep clause, target status, max Spikes, public Rapid Spin, Explosion blocking, and unavailable preservation pivots. Run `python -m tools.boss_ai_preference type-evidence` to generate the type-effectiveness firewall report at `audit/boss_ai_preference/type_effectiveness_evidence.md` and `audit/boss_ai_preference/type_effectiveness_evidence.json`. This report parses `data/types/type_matchups.asm`, checks the 15 documented romhack chart edits, and scans benchmark/oracle/policy explanations for words like `super-effective`, `resisted`, `immune`, and `neutral`. A claim must be tied to the environment-specific source paths before it is allowed. Run `python -m tools.boss_ai_preference long-battle-review` to generate the current structured 30+ turn review at `audit/boss_ai_preference/long_battle_review.md` and `audit/boss_ai_preference/long_battle_review.json`. This is a constructed vanilla-GSC review with 32 ledgered turns, critical-turn labels, earliest-error classification, and benchmark extraction targets. Run `python -m tools.boss_ai_preference benchmark-harvest` to inspect which existing preference fixtures can already be promoted into benchmark cards and which still lack a best / acceptable / catastrophic label piece. Run `python -m tools.boss_ai_preference benchmark-label-queue` to turn those partial harvested fixtures into concrete review requests instead of guessing the missing labels.

Required public-card fields:

- `split`: one of `seed`, `holdout`, or `fixture_harvest`; held-out cards are

for answer-changing variants that should not merely restate the seed case, while harvested cards are promoted from saved fixture preference feedback.

- `mechanics_profile`: either `vanilla_gsc` or `romhack_gym_leader_lab`; do

not mix vanilla conclusions with romhack mechanics.

- `position_snapshot`: visible battle state only.
- `candidate_moves_public`: legal/effective candidate move IDs without labels

or reasons.

- `required_answer_fields`: structured fields the policy answer must emit.
- `public_opponent_model`: the four required tiers are `immediate_punish`, `role_aware_preservation`, `incentive_compatible_response`, and `hidden_info_belief_state`.
- `hidden_info_visible_to_policy`: explicit public uncertainty, not oracle

truth.

Required hidden-oracle fields:

- `best`, `acceptable`, and `catastrophic`: hidden move labels used only by the

evaluator.

- `why_best`, `why_acceptable`, and `why_catastrophic`: expert rationale keyed

by move ID.

- `catastrophe_branches`: exact trigger and consequence for losing lines.
- `answer_changing_information`: facts that would flip or change the answer.
- `heuristic_arbitration`: competing heuristics, dominant policy, and

exceptions.

- `current_win_conditions` and `irreplaceable_pieces`: oracle route and

preservation targets used for review.

Type-effectiveness firewall:

- Keep chart result, passive-adjusted result, final damage, and strategic move

label as separate claims.

- Use `data/types/type_matchups.asm` for chart facts; this romhack already has

15 known chart edits, including Dark -> Steel neutral and Ghost -> Steel no effect.

- Use `engine/battle/type_passive_damage_mods.asm` for Dragon's Majesty,

Imperial Scales, Dark status shield, and other passive adjustments.

- Use validated debugger traces for exact battle-state damage, not as a blanket

override of source. Source/debugger disagreement means source-version or tooling mismatch and must be investigated.

- Do not copy vanilla Steel/Ghost/Dark examples into romhack benchmarks unless

the chart source is referenced and the romhack result is stated.

Initial hard snapshots:

Benchmark	Split	Profile	Arbitration lesson
vanilla_gsc_sleep_setup_disruption_001	seed	vanilla GSC	After sleep misses, re-score before deciding whether sleep, chip, or setup dominates.
romhack_spikes_third_layer_janine_001	seed	romhack	Finish the third Spikes layer only when the current state makes the layer live and hard to remove.
romhack_spikes_fourth_click_janine_001	seed	romhack	A maxed Spikes state makes another Spikes click a failed move, regardless of the generic hazard heuristic.
romhack_explosion_route_trade_brock_001	seed	romhack	Explosion is correct only when the sacrifice buys a named next position and does not delete an irreplaceable answer.
romhack_defensive_answer_preservation_pryce_001	seed	romhack	Preserve the bulky pivot by moving through the resistant answer; do not trust type-slogan shortcuts.
vanilla_gsc_sleep_clause_blocked_holdout_001	holdout	vanilla GSC	Sleep/setup must collapse into a live move when Sleep Clause blocks the control route.
romhack_spikes_public_spinner_holdout_001	holdout	romhack	Public hazard control can dominate finishing the third layer.
romhack_spikes_maxed_explosion_conversion_holdout_001	holdout	romhack	At max layers, Explosion can dominate live chip if it removes the hazard absorber.
romhack_explosion_blocked_by_protect_holdout_001	holdout	romhack	Revealed Protect can override the Explosion-converts heuristic.
romhack_defensive_answer_unavailable_holdout_001	holdout	romhack	If the defensive pivot is gone, choose a live mitigation move instead of an impossible preservation line.
long_battle_sleep_disruption_after_miss_001	holdout	vanilla GSC	A missed sleep move forces re-scoring; raw setup is catastrophic while sleep remains live only if clause and target status still allow it.
long_battle_rest_tempo_unforced_001	holdout	vanilla GSC	Rest at high HP can be tempo loss when it is not yet forced by role survival.

Benchmark	Split	Profile	Arbitration lesson
fixture_chuck_poliwrath_vs_pidgeotto_ice_punch_001	fixture_harvest	romhack	A live preservation switch can dominate visible coverage when the active closer is still needed.
fixture_janine_qwilfish_third_spikes_layer_001	fixture_harvest	romhack	Fixture feedback confirms third-layer Spikes dominates chip while public spin and immediate KO routes are absent.
fixture_mechanics_snorlax_full_hp_rest_status_fail_001	fixture_harvest	romhack	Mechanics legality dominates role slogans: full-HP Rest and awake Sleep Talk are failed state transitions.
fixture_brock_golem_vs_vaporeon_explosion_question_001	fixture_harvest	romhack	Explosion can narrowly dominate a preservation switch when a low-future-value piece buys a clean next position.
external_gsc_forretress_explosion_on_quagsire_001	holdout	vanilla GSC	Replay 861526 converts a spent low-HP Forretress into a Quagsire-removal Explosion, then tests the blocked-Explosion boundary with a Protect mutation.
external_gsc_vaporeon_vs_restdtalk_snorlax_001	holdout	vanilla GSC	Replay 861526 tests the GSC RestTalk branch: sleeping Snorlax can still call Rest, so immediate Surf pressure dominates greedy Growth unless RestTalk is absent.
external_gsc_golem_late_rapid_spin_001	holdout	vanilla GSC	Replay 861526 tests late-game hazard removal: Rapid Spin preserves future switch/phaze routes while Snorlax is likely to Rest; no-hazards mutation flips to Earthquake.
fixture_will_slowbro_vs_houndoom_fast_dark_001	fixture_harvest	romhack	Existing Will feedback tests fast public Dark pressure: preserve Slowbro through the healthy Houndoom pivot; if the pivot is unavailable, the mutation flips to Surf as the best forced stay-in attack.
fixture_koga_ariados_vs_typhlosion_fire_spikes_001	fixture_harvest	romhack	Koga Ariados into revealed Typhlosion Flamethrower tests Spikes/status arbitration: preserve the hazard lead through Tentacruel; if Tentacruel is unavailable, the mutation flips to Umbreon.

Benchmark	Split	Profile	Arbitration lesson
fixture_bugsy_scyther_vs_geodude_safe_swords_dance_001	fixture_harvest	romhack	Bugsy Scyther into Geodude tests the positive setup case: Swords Dance is correct while the worst plausible punish is survivable; if the punish becomes a KO, the mutation flips to Wing Attack.
fixture_bugsy_scyther_vs_quilava_fire_setup_001	fixture_harvest	romhack	Bugsy Scyther into Quilava tests setup under live Fire pressure: Swords Dance is correct while Scyther acts first and survives Ember; if Ember removes the setup route, the mutation flips to Wing Attack.
fixture_chuck_poliwrath_vs_alakazam_psychic_pivot_001	fixture_harvest	romhack	Chuck Poliwrath into faster Alakazam tests sleep-vs-preservation arbitration: verified Psychic immunity makes Umbreon best; if Umbreon is unavailable, the mutation flips to Hypnosis.
fixture_morty_haunter_vs_noctowl_sleep_line_001	fixture_harvest	romhack	Morty Haunter into Noctowl tests the positive sleep-control case: Hypnosis is best while Sleep Clause is free and the target is unstated; if Sleep Clause is occupied, the mutation flips to Night Shade.
fixture_falkner_pidgeotto_vs_geodude_scout_probe_001	fixture_harvest	romhack	Falkner Pidgeotto into Geodude tests one-probe policy: Sand Attack is best while Rock Throw is a live punish and Gust is tiny chip; if the punish branch is removed, the mutation flips to Gust.
fixture_pryce_cloyster_vs_quilava_fire_pivot_001	fixture_harvest	romhack	Pryce Cloyster into faster Quilava tests bad-matchup switching: preserve the low-HP Spikes/Explosion resource through Slowking while Flame Wheel KOs before progress; if Cloyster survives, the mutation flips to Surf.
fixture_whitney_miltank_vs_geodude_rollout_lock_001	fixture_harvest	romhack	Whitney Miltank into Geodude tests lock-move sequencing: Body Slam pressure comes before Rollout; if Geodude is paralyzed and the board is safe, the mutation flips to Rollout.

Benchmark	Split	Profile	Arbitration lesson
fixture_karen_crobat_vs_dragonite_toxic_clock_001	fixture_harvest	romhack	Karen Crobat into Dragonite tests Toxic-clock timing: start the clock while Crobat survives the public Outrage punish; if survival disappears, the mutation flips to Tyranitar.
fixture_koga_crobat_vs_alakazam_immediate_ko_001	fixture_harvest	romhack	Koga Crobat into Alakazam tests immediate-KO arbitration: Wing Attack beats Toxic while it removes Alakazam now; if the KO threshold disappears, the mutation flips to Umbreon.
fixture_jasmine_skarmory_vs_machoke_focus_energy_001	fixture_harvest	romhack	Jasmine Skarmory into Machoke tests phazing arbitration: Toxic beats reflexive Whirlwind while Focus Energy is low urgency; if a real attack-boost route appears, the mutation flips to Whirlwind.
fixture_morty_misdreavus_vs_typhlosion_perish_route_001	fixture_harvest	romhack	Morty Misdreavus into boosted Typhlosion tests forced-clock routing: Perish Song wins while Misdreavus survives Flame Wheel; if survival disappears, the mutation flips to Gengar.
fixture_whitney_clefairy_vs_bayleef_encore_reflect_001	fixture_harvest	romhack	Whitney Clefairy into Bayleef tests Encore timing: Encore wins when Reflect was the visible last move and Clefairy is faster; if the trap is absent, the mutation flips to Double Team.
fixture_jasmine_magneton_vs_quilava_speed_control_001	fixture_harvest	romhack	Jasmine Magneton into Quilava tests speed-control status: Thunder Wave wins while Thunderbolt cannot remove the faster threat; if the damage threshold reaches direct removal, the mutation flips to Thunderbolt.
fixture_misty_starmie_vs_meganium_recover_tempo_001	fixture_harvest	romhack	Misty Starmie into Meganium tests recovery timing: Psychic wins while Recover lacks a real window and chip enables Lapras cleanup; if Meganium cannot punish and the chip threshold is gone, the mutation flips to Recover.

Benchmark	Split	Profile	Arbitration lesson
fixture_clair_dragonair_vs_suicune_hidden_ice_001	fixture_harvest	romhack	Clair Dragonair into Suicune tests hidden-info preservation: Kingdra wins while hidden Ice coverage is plausible; if that punish is no longer plausible, the mutation flips to Thunder.
fixture_bugsy_ariados_vs_pidgey_status_clock_001	fixture_harvest	romhack	Bugsy Ariados into Pidgey tests status-clock sequencing: Toxic wins while the target is unstated and Ariados survives; if the clock already exists, the mutation flips to Scyther.

Key lesson: `set Spikes`, `preserve answers`, `use sleep`, and `Explosion converts` can all be true. The policy has to decide which one dominates on the exact turn, under the correct mechanics profile and opponent model.

Current fixture coverage audit:

- Latest generated reports:
 - `audit/boss_ai_preference/latest_report.md`
 - `audit/boss_ai_preference/lesson_report.md`
 - `audit/boss_ai_preference/active_queue.md`
 - `audit/boss_ai_preference/plan_queue.md`
 - `audit/boss_ai_preference/expert_play_research.md`
 - `audit/boss_ai_preference/feature_report.md`
- Current coverage is not close to mastery:
 - 57 total fixtures;
 - 0 direct action labels;
 - 52 pairwise preferences;
 - 27 / 57 fixtures with pairwise feedback;
 - 69 trajectory preferences and 3 plan demonstrations;
 - 42 machine-readable state-transition benchmarks: 5 seed, 15 holdout, and

22 fixture-harvest cards promoted from saved feedback, fixture evidence, or targeted state-transition gaps.

- 35 boundary mutation checks that force answer flips across sleep/setup,

maxed Spikes, public spin pressure, Explosion blocking, opening double-switch pressure, RestTalk branch removal, late-game hazard removal, phazing order, PP conservation, damage survival, defensive preservation availability, fast public-punish pivot availability, and speed-control versus direct-removal thresholds, recovery-window timing, and hidden-info coverage plausibility, and already-established status clocks.

- 15 romhack type-chart deltas enforced by the type-effectiveness evidence

report; current benchmark/oracle/policy text has 21 effectiveness claims and 0 unsupported claims.

- 19 source-backed expert-play principles in

`audit/boss_ai_preference/expert_play_research.md`, including the current type-slogan guard: effectiveness language must be tied to exact mechanics and a route-changing damage threshold, plus the Spikes guard that hazards are not a free passive turn under revealed lethal pressure, plus the setup guard that setup requires both a survivable punish branch and a route gain, plus the bad-matchup switching guard that a pivot can preserve future route resources when the active faints before progress.

- 132 extracted feature names in

`audit/boss_ai_preference/feature_report.md`; plan features now distinguish stay-in, setup, switch-preserve, and route-trade plans under public major/lethal active threats, with separate fast-threat markers.

- 1 structured 32-turn constructed long-battle review with 4 critical turns,

earliest irreversible error at turn 9, and 3 benchmark extraction targets.

- 4 complete fixture-derived benchmark candidates and 23 partial candidates

in `audit/boss_ai_preference/fixture_benchmark_harvest.md`; the complete candidates are now mirrored in the executable benchmark suite; the Will Slowbro/Houndoom fast-threat preservation case, the Koga Ariados/Typhlosion lethal Fire-pressure case, the Buggy Scyther/Geodude safe-setup case, the Chuck Poliwrath/Alakazam Psychic-pivot case, and the Morty Haunter/Noctowl sleep-control case, and the Falkner Pidgeotto/Geodude scout-probe case are added as targeted coverage cards with answer-flip mutations.

- 23 benchmark label-review requests in

`audit/boss_ai_preference/benchmark_label_queue.md`; 11 can become complete benchmark candidates with one non-conflicting acceptable-label review.

- Current reports identify active gaps where the scorer is uncertain or

lacks enough trajectory coverage. High-priority remaining gaps include

`falkner_pidgeotto_vs_geodude_rock_risk`, `pryce_cloyster_vs_quilava_explosion_line`, `koga_crobat_vs_alakazam_toxic_or_attack`. The trajectory model still has only one strict holdout row, even though the latest fit now predicts that row correctly. More non-holdout fast-threat and hazard-vs-preservation coverage is still needed before trusting that model family.

Coverage rule: do not claim the notebook can answer the lab broadly while direct labels are empty and most fixtures lack feedback. A benchmark card is study evidence, not a replacement for a saved preference or trajectory row.

Initial Benchmark Scenarios

Use these as the first regression checks. They are not a full mastery suite; they are the seed positions that turn the current notebook into something that can fail.

bench_sleep_setup_erika_001

- Fixture: `erika_victreebel_vs_snorlax_sleep_or_boost`
- Skill: sleep/setup branch control.
- Expected move class: Deny into Convert.
- Accepted line: Sleep Powder first, then one Swords Dance if sleep lands and

the board remains safe, then Sludge Bomb.

- Required explanation:
 - raw Sludge Bomb is not a KO at current HP;
 - raw setup risks taking heavy Body Slam chip first;
 - sleep is useful because it creates the setup turn that changes KO math;
 - after a miss, wake, switch, or Sleep Clause block, re-score instead of

continuing the script.

- Evidence:
 - damage debugger: unboosted Sludge Bomb into 84% Snorlax is 168-198 damage,

51-60% max HP, not a guaranteed KO;

- damage debugger: Snorlax Body Slam into 77% Victreebel is 116-137 damage,

57-67% max HP;

- damage debugger: +2 Sludge Bomb into Snorlax is 342-402 damage, 103-121%

max HP, guaranteed from full;

- scratch simulation: sleep plus one boost outperformed raw attack, raw

setup, and sleep plus two boosts in the controlled Victreebel/Snorlax policy test.

- Failure signs:
 - choosing Swords Dance first without accounting for Body Slam;
 - choosing two boosts because sleep landed once;
 - continuing setup after a miss.

bench_sleep_setup_lance_yanma_001

- Fixture: `lance_yanma_vs_lapras_sleep_powder_or_quiver_dance`

- Skill: sleep/setup branch control under lethal coverage.
- Expected move class: Deny into Convert.
- Accepted line: Sleep Powder first. If it lands and the board remains safe,

use the created turn for Quiver Dance or immediate Giga Drain depending on wake/switch risk; if it misses or Sleep Clause blocks it, re-score before any setup.

- Required explanation:
 - raw Giga Drain is not meaningful damage into full Lapras;
 - raw Quiver Dance loses to the public Ice Beam branch;
 - Kingdra is the safer preservation pivot into Ice, but switching gives up

Yanma's immediate disruption route;

- Focus Band is not a plan, only a low-probability emergency branch.
- Evidence:
 - fixture: Yanma level 46 at full HP, Focus Band, with revealed Giga Drain

into Lapras level 47 at full HP with revealed Surf and public Ice Beam;

- damage debugger: Yanma Giga Drain into Lapras is 32-38 damage, 15-18% max

HP;

- damage debugger: Lapras Ice Beam into Yanma is 148-174 damage, guaranteed

KO from full;

- damage debugger: Lapras Surf into Yanma is 23-28 damage, 18-22% max HP;
- damage debugger: Lapras Ice Beam into Kingdra is 39-46 damage, 26-31% max

HP.

- Failure signs:
 - choosing Quiver Dance first because Yanma is a setup-style disruption mon;
 - choosing Giga Drain because it is super-effective without checking the

tiny damage;

- relying on Focus Band as if it were a stable defensive resource.

bench_status_bugsy_ariados_001

- Fixture: **bugsy_ariados_vs_pidghey_toxic_or_attack**
- Skill: status tempo, weak-chip recognition, and ace preservation.
- Expected move class: Deny or Convert.

- Accepted line: Toxic is the best listed opener if the boss is allowed to

spend one turn creating a clock. Direct attacks are low-impact, and switching to Scyther exposes the ace before the board requires it.

- Required explanation:
 - Pidgey's public Flying pressure is real but not immediately lethal;
 - Ariados's direct damage into Pidgey is too small to win a damage race

quickly;

- Toxic changes future turns by forcing Pidgey to act under a clock or pivot;
- Scyther should not be thrown in just because it is the ace.
- Evidence:
 - fixture: Ariados level 15 at full HP, Berry, into level 17 Pidgey with

revealed Quick Attack and plausible Gust;

- damage debugger: Pidgey Gust into Ariados is 16-19 damage, 27-32% max HP;
- damage debugger: Pidgey Quick Attack into Ariados is 7-9 damage, 12-15%

max HP;

- damage debugger: Ariados Leech Life into Pidgey is 10-12 damage, 18-21%

max HP;

- damage debugger: Ariados Giga Drain into Pidgey is 10-12 damage, 18-21%

max HP.

- Failure signs:
 - switching to Scyther only because Pidgey is Flying-type;
 - choosing weak chip and calling it tempo when it does not change the next

turn;

- using Toxic without naming what the poison clock buys.
- Benchmark artifact: `fixture_bugsy_ariados_vs_pidgey_status_clock_001`;

mutation `mut_bugsy_status_clock_already_started_001` flips to Scyther when Pidgey is already toxic, because the status job is complete and the follow-up route must be re-scored.

`bench_switch_preserve_pryce_001`

- Fixture: `pryce_slowking_vs_ampharos_ground_pivot`

- Skill: switching, preservation, and public type pressure.
- Expected move class: Preserve or Deny.
- Accepted line: switch Slowking to Piloswine, then re-score from the new

board.

- Required explanation:
 - Slowking is not supposed to sit in front of the known Electric pressure;
 - Thunder Wave is slower than solving the immediate survival problem;
 - Rest donates tempo and does not answer the public threat;
 - Piloswine is not just a "Ground type" slogan; it materially changes the

damage race.

- Evidence:
 - damage debugger: Ampharos ThunderPunch into 68% Slowking is 75-89 damage,

68-81% max HP and a guaranteed KO at current HP;

- damage debugger: Ampharos ThunderPunch into Piloswine is 22-26 damage,

16-19% max HP;

- damage debugger: Piloswine Earthquake into 80% Ampharos is 76-90 damage,

52-62% max HP;

- damage debugger: Slowking Surf into 80% Ampharos is only 11-14 damage,

8-10% max HP.

- Failure signs:
 - clicking Thunder Wave because status is generally useful;
 - resting at high risk;
 - calling the switch good without checking that Piloswine actually absorbs

the public hit.

bench_cashout_brock_001

- Fixture: `brock_golem_vs_vaporeon_explosion_question`
- Skill: sacrifice and Explosion valuation.
- Expected move class: Cash out.
- Accepted line: Explosion is acceptable when the explanation names why Golem

is being spent and what the next entry gains. The answer is not "Explosion is dramatic"; it is "Surf already removes Golem, while Explosion converts the doomed active into removal of the current Water threat."

- Required explanation:
 - Vaporeon Surf guarantees the KO on Golem even from full;
 - Earthquake is not enough to stop Vaporeon;
 - Curse is too slow under public Water pressure;
 - switching to Omastar is a candidate, but preserving Golem must be worth

more than losing the chance to remove Vaporeon immediately.

- Evidence:
 - damage debugger: Vaporeon Surf into 42% Golem is 339-399 damage, 170-201%

max HP, guaranteed KO even from full;

- damage debugger: Golem Earthquake into 76% Vaporeon is 79-94 damage,

29-34% max HP and not a KO;

- damage debugger: Golem Explosion into 76% Vaporeon is 262-308 damage,

95-112% max HP and guaranteed KO at current HP.

- Failure signs:
 - preserving Golem without naming its future job;
 - exploding without naming the follow-up route;
 - using Curse into a known immediate KO.

bench_snorlax_tempo_red_001

- Fixture: red_snorlax_vs_alakazam_curse_or_body_slam
- Skill: tempo before setup, Snorlax route discipline.
- Status: provisional. Keep this as a study benchmark, not a stable rule,

until the rough threat report and direct damage debugger ranges are reconciled.

- Expected move class: Convert or Scout.
- Working accepted line: Body Slam before Curse, Rest, or awake Sleep Talk.
- Required explanation:
 - Curse may be strong later, but before paralysis or clearer pressure it can

donate a Recover/Psychic turn;

- Body Slam keeps tempo and can create paralysis, which may make later Curse

or direct pressure more real;

- Rest is too early at 84% HP;
- Sleep Talk is a public fail while awake.
- Current evidence:
 - damage debugger: Snorlax Body Slam into 62% Alakazam is 43-51 damage,

17-21% max HP and not a KO;

- damage debugger: Alakazam Psychic into 84% Snorlax is 133-157 damage,

39-47% max HP;

- threat availability report has a rougher Psychic estimate around 34-39%,

so do not promote the exact range until the mismatch is checked.

- Failure signs:
 - clicking Curse because Snorlax is a setup Pokemon;
 - using Rest or Sleep Talk without checking current state;
 - ignoring Alakazam's Recover and Psychic pressure.

bench_setup_tempo_will_001

- Fixture: will_slowbro_vs_houndoom_amnesia_or_surf
- Skill: setup tempo trap, speed-order awareness, and preservation.
- Expected move class: Preserve.
- Accepted line: switch Slowbro to Houndoom, then re-score.
- Required explanation:
 - Amnesia boosts the correct defensive side against Crunch, but Slowbro is

already slower and cannot spend that turn if Crunch is selected;

- Surf is the correct attacking lane into Houndoom only in branches where

Slowbro actually moves;

- Psychic is a public fail into Dark typing;
- the mirror Houndoom pivot gives up immediate Surf pressure but is the only

listed line that covers the revealed lethal Crunch branch.

- Evidence:
 - source fixture: Slowbro is level 41 at 78% HP with Leftovers; player

Houndoom is level 44 at 100% HP with revealed Crunch;

- local stat evidence: Houndoom's base Speed is 115 and Slowbro's is 30, so

Houndoom should move first absent priority or speed modifiers;

- damage debugger: Houndoom Crunch into 78% Slowbro is 170-200 damage,

113-132% max HP, guaranteed KO even from full;

- damage debugger: Slowbro Surf into Houndoom is 78-92 damage, 45-53% max HP,

but this does not matter in the stay-in Crunch branch because Slowbro faints first;

- damage debugger: Slowbro Psychic into Houndoom is 0 damage;
- damage debugger: Houndoom Crunch into the boss Houndoom pivot is 43-51

damage, 31-37% max HP.

- Failure signs:
 - choosing Amnesia because it boosts Special Defense without checking speed

and current KO range;

- choosing Surf as "clean direct damage" without naming the lethal faster

Crunch branch;

- switching only because of type matchup language, without checking that the

pivot actually survives the public hit.

bench_hazard_tempo_koga_001

- Fixture: koga_ariados_vs_typhlosion_spikes_or_toxic
- Skill: hazard greed, preservation, and switch-punish awareness.
- Expected move class: Preserve or Scout.
- Accepted line: switch Ariados to Tentacruel over Spikes, Toxic, or Leech

Life, then re-score immediately because Earthquake is a plausible punish.

- Required explanation:
 - three-layer Spikes are strategically powerful in this romhack, but the

setter still needs a playable turn;

- Ariados is slower than Typhlosion and cannot assume it gets to place

Spikes or Toxic under revealed Flamethrower pressure;

- Leech Life is resisted low-impact damage and does not solve the threat;
- Tentacruel covers the revealed Flamethrower branch, but the line is not

free because public notes say Earthquake is plausible.

- Evidence:
 - source fixture: Ariados level 40 at full HP, hazard lead; player Typhlosion

level 45 at full HP with revealed Flamethrower and plausible Earthquake;

- local stat evidence: Typhlosion's base Speed is 100 and Ariados's is 40,

so Typhlosion should move first absent priority or speed modifiers;

- damage debugger: Typhlosion Flamethrower into Ariados is 260-306 damage,

181-213% max HP, guaranteed KO;

- damage debugger: Ariados Leech Life into Typhlosion is 14-17 damage,

9-11% max HP;

- damage debugger: Typhlosion Flamethrower into Tentacruel is 34-40 damage,

28-33% max HP;

- damage debugger: Typhlosion Earthquake into Tentacruel is 141-166 damage,

118-138% max HP, guaranteed KO.

- Failure signs:
 - clicking Spikes because hazards are generally important while ignoring the

faster lethal revealed move;

- switching to Tentacruel without naming Earthquake as the worst plausible

punish;

- treating Toxic or Leech Life as progress when the active matchup may end

before Ariados moves.

bench_romhack_spikes_third_layer_janine_001

- Fixture: **janine_qwilfish_finish_third_spikes_layer**
- Skill: romhack hazard-layer valuation.
- Expected move class: Finish the hazard stack.
- Accepted line: use Spikes over ordinary Sludge Bomb, Surf, or switching when

two player-side layers are already down, the seen player team is grounded, no Rapid Spin user has been revealed, and no immediate KO threat is public.

- Required explanation:
 - this is not vanilla GSC one-layer Spikes;
 - the third local layer changes grounded switch-ins from max HP / 6 to max

HP / 4;

- the value comes from future switches, Rest cycles, phazing, and forced re-entries, not from the Spikes turn dealing damage immediately;

• Explosion remains a route-trade question, not a strict "always worse" alternative.

- Evidence:
 - source fixture: Janine Qwilfish level 60 with Spikes, Sludge Bomb, Surf, and Explosion;

- local mechanics docs and source: three layers are supported, with damage at /8, /6, and /4;

- strict labels prefer Spikes over Sludge Bomb, Surf, and switching, while the Explosion comparison is intentionally non-strict.

- Failure signs:
 - saying "Spikes are good" without naming current layer count;
 - treating the third layer like vanilla GSC chip;
 - cashing ordinary attack damage without explaining why the completed hazard stack is no longer needed.

bench_romhack_spikes_maxed_janine_001

- Fixture: janine_qwilfish_spikes_already_maxed
- Skill: hazard-state legality and redundancy.
- Expected move class: Use a live move; do not click Spikes.
- Accepted line: Sludge Bomb, Surf, Explosion, or a justified switch can all be live candidates, but Spikes itself is a failed fourth-layer attempt.

- Required explanation:
 - both local Spikes bits are already set for the player side;
 - the move-effect source fails placement at three layers;
 - once the stack is complete, Qwilfish should convert the hazard position

through damage, preservation, or a later Explosion route.

- Evidence:
 - strict labels prefer Sludge Bomb and Surf over Spikes;
 - scorer feature `spikes_already_maxed` is present for the failed move.
- Failure signs:
 - treating Spikes as useful because hazards are generally valuable;
 - failing to check the current stack before choosing a hazard move.

`bench_two_turn_red_charizard_001`

- Fixture: `red_charizard_vs_starmie_sunny_day_or_switch`
- Skill: two-turn plan discipline and candidate-set skepticism.
- Status: unresolved. Do not promote the fixture baseline as correct until the

candidate set is expanded or the opponent model explicitly says Starmie is unlikely to click revealed Surf.

- Expected move class: Preserve, but only if the switch actually survives the

revealed punish.

- Working verdict: all listed stay-in moves fail against revealed Surf, and

the listed Pikachu switch also fails if Starmie clicks Surf. The right review answer is "the shown candidates are incomplete or require a prediction assumption", not "Sunny Day is clever".

- Required explanation:
 - Starmie outspeeds Charizard;
 - Sunny Day plus SolarBeam is a two-turn package from neutral weather and

does not solve the immediate Surf branch;

- SolarBeam without sun charges and is therefore even slower;
- Flamethrower and Wing Attack do not threaten enough damage;
- Pikachu threatens Starmie if it gets in safely, but it does not absorb Surf

at 64% HP.

- Evidence:
 - source fixture: Charizard level 77 at 92% HP, Charcoal, neutral weather;

Starmie level 80 at 100% HP with revealed Surf;

- local stat evidence: Starmie and Charizard both have 115 base Speed in this

hack, but the fixture states Starmie outspeeds;

- damage debugger: Starmie Surf into 92% Charizard is 262-308 damage,

108-127% max HP, guaranteed KO even from full;

- damage debugger: Charizard Flamethrower into Starmie is 39-46 damage,

14-17% max HP;

- damage debugger: Charizard Wing Attack into Starmie is 57-67 damage,

21-24% max HP;

- damage debugger: Starmie Surf into 64% level 81 Pikachu is 274-322 damage,

136-160% max HP, guaranteed KO even from full;

- damage debugger: Light Ball Pikachu Thunderbolt into Starmie is 145-171

damage, 52-62% max HP, not a guaranteed KO from full.

- Failure signs:
 - recommending Sunny Day because it enables SolarBeam without checking the

first turn;

- recommending switch Pikachu as if threat alignment equals safe entry;
- treating the listed action set as complete when every shown line loses to

the public punish.

bench_choice_lock_sabrina_001

- Fixture: **sabrina_alakazam_choice_lock_vs_houndoom**
- Skill: choice-lock discipline, Dark immunity, and coverage triage.
- Expected move class: Coverage or Preserve.
- Working accepted line: do not lock Choice Specs Alakazam into Psychic

against public Houndoom. ThunderPunch is the best listed direct coverage checked so far, but switching to Hypno remains a candidate if preserving the ace matters more than chip.

- Required explanation:
 - Psychic is a public fail into Dark typing, even with Choice Specs;
 - Houndoom's revealed Crunch threatens a guaranteed KO on the current

Alakazam HP, so the boss is not free to throw a low-impact move;

- coverage should be judged by concrete damage, not by "neutral is fine".
- Evidence:

- fixture: Alakazam level 67 at 89% HP, Choice Specs, into Houndoom level 66

at 72% HP with revealed Crunch and Flamethrower;

- damage debugger: Choice Specs Psychic into Houndoom is 0 damage;
- damage debugger: Choice Specs ThunderPunch into 72% Houndoom is 74-87

damage, 40-47% current HP;

- damage debugger: Choice Specs Ice Punch into 72% Houndoom is 36-43 damage,

19-23% current HP;

- damage debugger: Houndoom Crunch into 89% Alakazam is 158-186 damage,

guaranteed KO at current HP.

- Failure signs:
 - choosing Psychic because it is Alakazam's identity move;
 - treating "coverage" as solved without comparing ThunderPunch and Ice Punch;
 - ignoring Choice lock when choosing a low-payoff neutral move.

bench_priority_red_pikachu_001

- Fixture: red_pikachu_vs_dragonite_thunderbolt_or_extremespeed
- Skill: priority discipline, Dragon passive transfer, and speed-value checks.
- Expected move class: Convert.
- Working accepted line: Thunderbolt is still the best listed first hit because

Pikachu already outspeeds and ExtremeSpeed's priority does not add value.

- Required explanation:
 - priority is useful only when it changes move order or secures a range;
 - Light Ball special pressure is better than ExtremeSpeed here, but neither

move comes close to a KO;

- Dragonite's Outrage is lethal if Dragonite gets its attack turn, so the

move must be understood as best available damage, not a complete answer.

- Evidence:
 - fixture: Red Pikachu level 81, Light Ball, full HP, into player Dragonite

level 80, full HP, revealed Outrage;

- fixture note: Pikachu outspeeds Dragonite;
- damage debugger: Light Ball Thunderbolt into Dragonite is 44-52 damage,

13-15% max HP;

- damage debugger: Light Ball ExtremeSpeed into Dragonite is 25-30 damage,

7-9% max HP;

- damage debugger: Light Ball Surf into Dragonite is 14-17 damage, 4-5% max

HP;

- damage debugger: Dragonite Outrage into Pikachu is 381-448 damage,

guaranteed KO from full.

- Failure signs:
 - clicking ExtremeSpeed because priority sounds safer when the boss already

moves first;

- claiming Thunderbolt "solves" Dragonite instead of saying it is merely the

best listed damage;

- ignoring Dragon-specific damage behavior and local damage ranges.

bench_mirror_coat_red_blastoise_001

- Fixture: red_blastoise_vs_zapdos_mirror_coat_or_blizzard
- Skill: reactive-counter survival checks and branch labeling.
- Status: provisional. Mirror Coat is a real high-upside line only on the

branches where Blastoise survives the revealed Electric attack.

- Expected move class: Counterplay or Preserve, depending on risk tolerance and

whether the battle state requires a swing.

- Working accepted line: Mirror Coat must be explained as a survival-roll trap,

not as a guaranteed answer. If the line cannot tolerate being KOed by Thunderbolt, switch Snorlax or another special sponge must be considered.

- Required explanation:
 - Mirror Coat does nothing if Blastoise faints first;
 - Zapdos Thunderbolt is a possible KO at current HP, so this is not a safe

baseline;

- Blizzard hits Zapdos but is not enough damage to solve the matchup by

itself.

- Evidence:
 - fixture: Blastoise level 77 at 88% HP, Mystic Water, into Zapdos level 78

with revealed Thunderbolt;

- damage debugger: Zapdos Thunderbolt into 88% Blastoise is 210-247 damage, 98-115% current HP, possible KO;
- damage debugger: Blastoise Blizzard into Zapdos is 112-132 damage, 37-44%

max HP;

- reactive implication: surviving Thunderbolt would let Mirror Coat return

more than Zapdos's full HP, but the survival branch is not guaranteed.

- Failure signs:
 - recommending Mirror Coat without first checking survival;
 - recommending Surf or Earthquake as if they address Zapdos;
 - treating a possible-KO branch as safe because the counterplay is stylish.

bench_prediction_karen_houndoom_001

- Fixture: karen_houndoom_vs_alakazam_pursuit_or_flamethrower
- Skill: prediction discipline and stay/switch branch comparison.
- Expected move class: Predict or Convert.
- Working accepted line: Pursuit is fair only if the explanation names why

Alakazam is likely to switch or why the switch branch is worth the weaker stay-in damage. If judging without that read, Crunch is the stronger direct pressure into a staying Alakazam.

- Required explanation:
 - Houndoom threatens Alakazam heavily, so a switch is plausible public

information;

- Pursuit has a high-payoff switch branch but is weak if Alakazam stays;
- Crunch is better direct damage into the stay branch than Flamethrower or

non-doubled Pursuit.

- Evidence:
 - fixture: Karen Houndoom level 47, Black Glasses, into player Alakazam

level 47 at full HP with revealed Psychic;

- damage debugger: Houndoom Pursuit into staying Alakazam is 42-50 damage,

28-34% max HP;

- damage debugger: Houndoom Flamethrower into Alakazam is 41-49 damage,

28-33% max HP;

- damage debugger: Houndoom Crunch into Alakazam is 91-108 damage, 61-72%

max HP;

- fixture note: Pursuit doubles damage on a switching target.
- Failure signs:
 - choosing Pursuit because it sounds advanced without naming the switch

evidence;

- choosing Flamethrower as "safe" while ignoring Crunch's much better stay

branch;

- switching away from Houndoom without naming the threat that forces it.

bench_status_karen_crobat_001

- Fixture: karen_crobat_vs_dragonite_toxic_or_attack
- Skill: status clock into lock-in, anti-Dragon pivot timing.
- Expected move class: Deny into Preserve.
- Working accepted line: Toxic is the best first listed move if Crobat can

survive the initial Outrage branch, then the boss should re-score whether to pivot Tyranitar, use Confuse Ray, or preserve Crobat. Immediate Wing Attack is too small to matter.

- Required explanation:
 - Dragonite's Outrage lock makes a poison clock meaningful;
 - Crobat survives one non-critical Outrage from current HP but cannot ignore

the follow-up turns;

- Tyranitar is a real anti-Dragon pivot, but switching immediately spends the

chance to start the clock;

- Confuse Ray is swingy and should not replace the more deterministic Toxic

clock without a reason.

- Evidence:
 - fixture: Karen Crobat level 45 at 92% HP, Leftovers, into player Dragonite

level 48 at full HP with revealed Outrage;

- damage debugger: Crobat Wing Attack into Dragonite is 25-30 damage, 12-14%

max HP;

- damage debugger: Dragonite Outrage into 92% Crobat is 87-103 damage,

63-74% current HP;

- damage debugger: Dragonite Outrage into Tyranitar is 84-99 damage, 55-65%

max HP;

- damage debugger: Tyranitar Rock Slide into Dragonite is 81-96 damage,

39-46% max HP.

- Failure signs:
 - clicking Wing Attack because it is STAB without checking damage impact;
 - switching Tyranitar immediately without comparing the Toxic-first branch;
 - leaving Crobat in for repeated Outrage turns after the clock has been set.
- Benchmark artifact: `fixture_karen_crobat_vs_dragonite_toxic_clock_001`;

mutation `mut_karen_toxic_clock_no_survival_001` flips to Tyranitar when Crobat no longer survives the immediate Outrage punish.

`bench_cashout_bruno_onix_001`

- Fixture: `bruno_onix_vs_typhlosion_sandstorm_or_explosion`
- Skill: low-HP cashout versus slow weather value.
- Expected move class: Cash out.
- Working accepted line: Explosion is the clean conversion if the boss accepts

spending Onix. Sandstorm is too slow as the main line because Earthquake removes Onix immediately and the weather chip does not remove Typhlosion.

- Required explanation:
 - Onix is already in guaranteed Earthquake KO range;
 - Rock Slide does not KO Typhlosion from current HP;
 - Explosion does KO Typhlosion from current HP;
 - Sandstorm is only worth considering if the downstream team specifically

benefits from the weather after Onix dies.

- Evidence:
 - fixture: Onix level 43 at 31% HP, Hard Stone, into Typhlosion level 45 at

61% HP with revealed Flamethrower and Earthquake;

- damage debugger: Typhlosion Earthquake into 31% Onix is 57-68 damage,

guaranteed KO at current HP;

- damage debugger: Onix Rock Slide into 61% Typhlosion is 66-78 damage,

69-81% current HP and not a KO;

- damage debugger: Onix Explosion into 61% Typhlosion is 120-142 damage,

guaranteed KO at current HP.

- Failure signs:
 - choosing Sandstorm because it creates long-term chip without checking

whether Onix survives or the chip changes the next turn;

- choosing Rock Slide because it is super-effective while missing the

non-KO into guaranteed Earthquake punish;

- preserving Onix without naming a future job that beats removing

Typhlosion now.

bench_mirror_coat_negative_red_blastoise_001

- Fixture: **red_blastoise_vs_tyranitar_surf_or_mirror_coat**
- Skill: reactive-counter negative example and simple pressure.
- Expected move class: Convert.
- Working accepted line: Surf is the right listed pressure. Mirror Coat is not

a good default because the opponent has obvious physical branches and Surf already hits Tyranitar's weakness.

- Required explanation:
 - Mirror Coat only answers special damage, and Tyranitar's public profile

includes Rock Slide plus plausible Earthquake;

- Crunch is special, but it is not the only or most defining punish branch;
- Surf is super-effective STAB and is stronger than Blastoise Earthquake.

- Evidence:
 - fixture: Red Blastoise level 77 at 76% HP, Mystic Water, into player

Tyranitar level 78 at 82% HP with revealed Rock Slide and Crunch;

- damage debugger: Blastoise Surf into 82% Tyranitar is 114-134 damage,

46-54% current HP;

- damage debugger: Blastoise Earthquake into 82% Tyranitar is 79-94 damage,

32-38% current HP;

- damage debugger: Tyranitar Rock Slide into 76% Blastoise is 91-108 damage,

49-58% current HP;

- damage debugger: Tyranitar Earthquake into 76% Blastoise is 80-95 damage,

43-51% current HP;

- damage debugger: Tyranitar Crunch into 76% Blastoise is 77-91 damage,

42-49% current HP.

- Failure signs:
 - choosing Mirror Coat because it was good in the Zapdos fixture;
 - treating Crunch's special category as enough reason to ignore Rock Slide

and Earthquake;

- using Earthquake because Rock/Dark sounds physical while Surf is stronger.

bench_delta_lance_dragonite_001

- Fixture: lance_dragonite_vs_aerodactyl_outrage_or_dragon_dance
- Skill: mechanics-source conflict handling.
- Status: active benchmark with a fixture-note caveat.
- Working accepted line: Outrage is the clean listed pressure, but the answer

must mention lock-in risk and the seen Vaporeon / Zapdos follow-ups. Dragon Dance is greedy under faster Rock Slide pressure unless the damage branch is explicitly acceptable.

- Evidence:
 - fixture note: Aerodactyl outspeeds Dragonite slightly and Outrage is clean

pressure with lock-in risk;

- source chart: data/types/type_matchups.asm has `GROUND, FLYING,

NO_EFFECT`;

- caveat: the fixture note "Earthquake cannot hit a Flying target" is too

broad for Dragonite because Dragon's Majesty converts immunities into resistances for damaging non-fixed moves by Dragon-typed attackers;

- control checks: non-Dragon Onix / Donphan / Golem Earthquake into Flying

targets returned 0 damage;

- damage debugger: Dragonite Outrage into Aerodactyl is 68-81 damage, 34-41%

max HP;

- damage debugger: Dragonite Earthquake into Aerodactyl is 45-54 damage,

23-27% max HP, which is weaker than Outrage despite bypassing the usual immunity;

- damage debugger: Aerodactyl Rock Slide into 92% Dragonite is 119-140

damage, 69-81% current HP and not a KO.

- Failure signs:
 - repeating the broad "Ground cannot hit Flying" rule without checking

Dragon's Majesty;

- clicking Dragon Dance without checking faster Rock Slide pressure and

lock-in alternatives;

- ignoring seen Vaporeon and Zapdos when evaluating Outrage lock.

bench_external_belly_drum_wake_001

- Source: external GSC OU replay,

<https://replay.pokemonshowdown.com/gen2ou-2404705604-gl00guvgm8l6etle1dd06927fn0jlsbw.log>.

- Turns: 44-46.
- Skill: sleep counter tracking, setup discipline, and preserved-answer checks.
- Expected move class: Reject Greedy Setup.
- Accepted answer: do not treat Belly Drum as safe merely because opposing

Snorlax is currently asleep on the bench. The correct review must say that the opposing Snorlax can return near full, wake or act before the setup route converts, and KO the Drummed Snorlax after Belly Drum cuts HP.

- Required explanation:
 - p2 Snorlax re-entered at 88% after Spikes and healed to 99% while asleep;

- p1 Snorlax re-entered at 63% and healed to 69%;
- Belly Drum dropped p1 Snorlax to 19%;
- p2 Snorlax then used Double-Edge and KOed it;
- the earlier Skarmory answer had been weakened but not enough to make the

Snorlax setup route automatically safe.

- Evidence:
 - replay ledger tags turn 46 as setup plus faint;
 - external review showed the same player tried Belly Drum twice, and both

attempts failed because the opponent preserved or recovered the answer in time.

- Failure signs:
 - saying "Snorlax is asleep, so this is the setup window" without tracking

wake or action timing;

- ignoring the HP cost of Belly Drum;
- reviewing only the final Double-Edge instead of the prior route map.

bench_external_phaze_loop_golem_skarm_001

- Source: external GSC OU replay,

<https://replay.pokemonshowdown.com/smogtours-gen2ou-505883.log>.

- Turns: 32-68, with key checkpoints at 36, 39, 49, 52, 59, and 68.
- Skill: phazing-loop review, forced-entry accounting, and endgame conversion.
- Expected move class: Long-game ledger, not one-turn move choice.
- Accepted answer: identify that Golem's Roar plus Earthquake was not random

stalling. It repeatedly denied Skarmory's Curse plan, forced Snorlax entries, and controlled which endgame pieces could appear. Skarmory's Whirlwind on turn 59 mattered because it dragged Starmie in; Golem's Explosion on turn 68 then removed or crippled Skarmory enough for Starmie to finish with Surf and Nightmare pressure into sleeping Snorlax.

- Required explanation:
 - name both sides' routes during the loop;
 - track which forced entries take damage or burn sleep turns;
 - distinguish Roar as progress from Roar as empty repetition;
 - explain why the final Golem Explosion was a route trade, not isolated

damage;

- note how Starmie's final entry converted the loop into a win.

- Evidence:
 - replay ledger tags repeated Golem Roar turns as phaze events;
 - turn 59: Golem's Roar failed while Skarmory's Whirlwind dragged Starmie;
 - turn 68: Golem Exploded into Skarmory;
 - turns 69-71: Starmie used Surf / Nightmare / Surf to end the game.
- Failure signs:
 - summarizing turns 32-68 as "they switched and phased a lot";
 - claiming Skarmory was winning because it kept using Curse;
 - missing that the loop was setting up Starmie's final route.

bench_external_restitalk_zapdos_001

- Source: external GSC OU replay,

<https://replay.pokemonshowdown.com/smogtours-gen2ou-912130.log>.

- Turns: 24-28.
- Skill: RestTalk branch accounting and conversion review.
- Expected move class: Continue Pressure With Branch Awareness.
- Accepted answer: pressuring the sleeping Zapdos can be correct, but the

explanation must not call the route converted while Sleep Talk can still call Rest. In the replay, Zapdos Rested on turn 24, was pushed to 27% on turn 25, then Sleep Talk called Rest and reset to full. The line only remains good if the attacking side can continue making progress after that reset.

- Required explanation:
 - p2 Zapdos used Rest on turn 24 and p1 Zapdos Thunder put it at 55%;
 - p1 Zapdos Hidden Power dropped it to 27% on turn 25;
 - p2 Zapdos Sleep Talk called Rest and restored to full;
 - Sleep Talk Rest is a legal branch to price, not a post-hoc excuse;
 - future Zapdos pressure must be re-scored after the reset.
- Evidence:
 - replay ledger tags turn 24 as recovery and turn 25 as Sleep Talk;
 - GSC mechanics allow Sleep Talk to call Rest and reset the Rest cycle.
- Failure signs:
 - saying "Zapdos is asleep, so finish it" without listing Sleep Talk

branches;

- treating Sleep Talk Rest as pure luck rather than a known legal branch;
- ignoring that the HP reset changes later Thunder and Hidden Power trades.

bench_external_spin_relayer_cloyster_001

- Source: external GSC OU replay,

<https://replay.pokemonshowdown.com/smogtours-gen2ou-912130.log>.

- Turns: 33 and 40-43.
- Skill: hazard-control durability and sacrifice valuation.
- Expected move class: Temporary Control, Then Re-check Spiker Access.
- Accepted answer: Golem's Rapid Spin on turn 33 removed Spikes from don

marto's side, but it did not permanently solve the hazard game because leoperi still had Cloyster. The poisoned Cloyster re-entered, set Spikes on turn 41, Exploded into the opposing Cloyster on turn 42, and let Houndoom remove it on turn 43. The sacrifice is good only because it restored switch tax and removed opposing hazard control / Water counterplay.

- Required explanation:
 - name the spinner, the remaining Spiker, and how the Spiker re-entered;
 - track Golem's HP after Spikes and Rhydon pressure;
 - explain why low-HP Cloyster still had one high-value job;
 - judge Explosion by the route it opened, not by immediate damage alone;
 - identify which future switches were taxed after Spikes returned.
- Evidence:
 - replay ledger tags turn 33 as Rapid Spin and turn 41 as Spikes;
 - turn 42 is a sacrifice event and turn 43 removes the opposing Cloyster.
- Failure signs:
 - saying "I spun, so hazards are solved";
 - ignoring the opponent's last chance to re-layer;
 - judging Explosion only by whether it immediately KOed the target.

bench_external_sleeping_lax_endgame_001

- Source: external GSC OU replay,

<https://replay.pokemonshowdown.com/smogtours-gen2ou-912130.log>.

- Turns: 1-2 and 43-51.
- Skill: sleeping win-condition tracking, wake timing, and endgame route audit.
- Expected move class: Preserve But Do Not Overcount Sleeping Route.
- Accepted answer: don marto preserved the Snorlax that opened with Curse and

was put to sleep by Lovely Kiss, but preservation did not make it immediately available. When it returned late, it was still asleep, took Spikes and Leech Seed pressure, faced Curse Rhydon, woke on turn 51, and died after one Earthquake. The route had to be evaluated by wake timing and entry damage, not by the label "Snorlax saved for late game."

- Required explanation:
 - the early Curse did not matter until Snorlax could act again;
 - carry sleeping bench state forward instead of treating the switch-out as a

reset;

- hazards and Leech Seed changed the wake-turn math;
- Rhydon's Curse on turn 49 converted the sleep window into an endgame route;
- the final route belonged to leoperi's Zapdos after Rhydon traded through

Snorlax and Zapdos.

- Evidence:
 - replay ledger shows Lovely Kiss on turn 2 and sleeping Snorlax returning on

turns 43-51;

- turn 49 is a setup event for Rhydon while Snorlax remains asleep;
- turns 51-54 show the Rhydon trade opening Zapdos's final Thunder route.
- Failure signs:
 - saying "Snorlax was saved for late game" without sleep and wake timing;
 - ignoring that asleep Snorlax cannot stop Rhydon setup immediately;
 - reviewing only turn 51 instead of the early Lovely Kiss and hazard pressure.

Romhack Fixture Check: Erika Sleep Setup

Fixture: `erika_victreebel_vs_snorlax_sleep_or_boost`.

Source state:

- Erika Victreebel level 64, 77% HP, Muscle Band, revealed Razor Leaf and

Sludge Bomb.

- Player Snorlax level 65, 84% HP, revealed Body Slam and Rest, with Curse and

Earthquake plausible.

Romhack damage debugger checks:

Query	Result
Victreebel Sludge Bomb into 84% Snorlax	168-198 damage, 51-60% max HP; not a guaranteed KO.
Snorlax Body Slam into 77% Victreebel	116-137 damage, 57-67% max HP; not guaranteed, but leaves Victreebel badly chipped.
+2 Victreebel Sludge Bomb into Snorlax	342-402 damage, 103-121% max HP; guaranteed KO even from full.

Interpretation: raw Sludge Bomb is pressure but not closure, and raw Swords Dance risks taking a huge Body Slam first. Sleep Powder is valuable because it can create the one safe Swords Dance turn that converts Sludge Bomb from chip into a guaranteed KO. The plan still branches: if Sleep Powder misses, do not pretend the setup turn was earned.

Current Heuristics

Immediate KO Beats Slow Play

- Status: stable
- Type: weight hint
- Rule: If a move reliably KO's or nearly KO's the active target now, prefer it

over slow status, passive recovery, weak chip, or greedy setup.

- Applies when: the boss can take the KO without exposing a more valuable plan.
- Does not apply when: the KO move creates a worse forced lock, misses a better

safe switch, or the target has a known public countermeasure.

- Example fixtures:
 - koga_crobat_vs_alakazam_toxic_or_attack
 - morty_gengar_vs_kadabra_destiny_bond
 - misty_starmie_vs_meganium_recover_or_attack
- Benchmark artifact: fixture_koga_crobat_vs_alakazam_immediate_ko_001;

mutation mut_koga_ko_removed_preserve_umbreon_001 flips to Umbreon when Wing Attack no longer removes Alakazam before Psychic or Recover pressure.

- Implementation target: debugger scoring first; ROM move scoring after review.

Phazing Needs A Live Setup Route

- Status: provisional
- Type: arbitration policy
- Rule: Whirlwind/Roar should dominate when the opponent has a real boost or

forced-route threat that must be denied now. Do not spend phazing just because the opponent used a setup-looking move.

- Applies when: the setup changes damage, makes the target uncontainable, burns

scarce defensive answers, or combines with hazards into progress.

- Does not apply when: the public setup signal is low urgency, the phazer has

scarce PP, status creates a better clock, or phazing fails under Gen 2 move order.

- Benchmark artifact: `fixture_jasmine_skarmory_vs_machoke_focus_energy_001`;

mutation `mut_jasmine_real_boost_requires_whirlwind_001` flips to Whirlwind when Machoke's setup route becomes a real attack-boost threat.

- Source note: Smogon's GSC phazing material supports phazing as route denial

and hazard pressure, while the move-priority article keeps Gen 2 timing as a separate mechanics gate.

Encore Needs A Public Trap Target

- Status: provisional
- Type: sequence policy
- Rule: Encore is a route move only when public last-move, speed, and fail-state

evidence show it can lock the target into a low-value action. Without that evidence, treat it as speculative utility.

- Applies when: the target just used a support, recovery, Protect-style, or

already-active screen move, the user acts first, and Encore is not already active or blocked by its local fail gates.

- Does not apply when: the last move is unknown, damaging, Struggle, Encore,

Mirror Move, the user is slower, or the target can switch freely and the trap does not create a route.

- Benchmark artifact: `fixture_whitney_clefairy_vs_bayleef_encore_reflect_001`;

mutation `mut_whitney_encore_trap_absent_double_team_001` flips to Double Team when Bayleef's visible last move is no longer Reflect.

- Source note: local mechanics show Encore lasts 3-6 turns and fails without a

valid last move; the play-policy lesson is to use that only when it changes the transition, not because Encore is generically clever.

Speed Control Must Buy A Future Turn

- Status: provisional
- Type: arbitration policy

- Rule: paralysis or speed-control status is strongest when it changes who acts

first on the next meaningful turn and the user survives the public punish. If direct damage already removes the active threat, attack instead.

- Applies when: the target is unstated, the speed flip changes the next

action, direct damage does not remove the target, and the status user survives the immediate branch.

- Does not apply when: the target is already stated, the direct attack reaches

a removal threshold, the status user faints before the speed control matters, or the revealed/implied counterplay makes the next turn worse than attacking.

- Benchmark artifact: `fixture_jasmine_magneton_vs_quilava_speed_control_001`;

mutation `mut_jasmine_speed_control_to_direct_ko_001` flips to Thunderbolt when the public damage threshold removes Quilava.

- Source note: Smogon's prediction and risk/reward guides frame play as public

information plus branch cost. Here that means Thunder Wave is not a slogan; it is a state transition that must beat the direct-removal branch.

Reliable Recovery Needs A Window

- Status: provisional
- Type: recovery / tempo policy
- Rule: Recover-style healing is correct only when it preserves a needed route

and the opponent cannot immediately erase the healing turn or improve a stronger route. If attacking changes a cleanup threshold, attack first.

- Applies when: the user is needed later, the opponent cannot punish the

recovery turn, and healing changes a future switch, KO, or endgame route.

- Does not apply when: recovery only loops the same losing damage exchange, a

direct attack creates a cleanup threshold, or the opponent can use the free turn for recovery, setup, hazards, or a decisive switch.

- Benchmark artifact: `fixture_misty_starmie_vs_meganium_recover_tempo_001`;

mutation `mut_misty_recovery_window_opens_001` flips to Recover when Meganium cannot punish and Psychic no longer changes the Lapras cleanup band.

- Source note: Smogon risk/reward framing and progress-oriented play both

support evaluating recovery by branch cost and route change, not by HP alone.

State-Invalid Moves Are Not Lines

- Status: stable
- Type: hard rule
- Rule: Reject moves whose current-state precondition is false before scoring

taste, damage, setup, or style.

- Applies when: Sleep Talk is chosen while awake, a sleep move is blocked by

Sleep Clause or existing status, status is repeated into an already-statused or immune target, Rapid Spin has no hazard-removal job and no meaningful chip target, a charge/setup package cannot survive its first turn, or recovery is unavailable or meaningless in the current state.

- Does not apply when: the move has a secondary legal purpose in the current

state and that purpose is explicitly named.

- Example evidence:
 - long-battle drill 2: awake Sleep Talk wasted turns for Snorlax and Raikou.
 - long-battle drill 2: Steelix repeatedly clicked Toxic into an already badly

poisoned Machop.

- long-battle drill 3: Cloyster clicked Toxic into Steel-type Forretress, and

Starmie clicked Rapid Spin with no hazards to remove.

- local Rest source check: Rest at exactly full HP jumps to the failed-move

path before the local engine reaches status-cure or sleep-counter code.

- `mechanics_snorlax_full_hp_rest_status_fail`: full-HP Rest and awake Sleep

Talk are both saved as strict negative pairwise evidence.

- Implementation target: pre-scoring fail gates and state preconditions.

Rest Needs A Real Recovery Window

- Status: stable
- Type: hard rule / resource policy
- Rule: Direct Rest is legal only if the user is below full HP. It is good only

when the HP/status recovery is worth the donated sleep turns or Sleep Talk turns.

- Applies when: the user is damaged enough that Rest succeeds, the restored HP

preserves an important route, and the opponent cannot convert the sleep turns into a decisive setup, phazing, trap, hazard, or Explosion branch.

- Does not apply when: the user is at exactly full HP, the user is healthy and

unstatused, the opponent has an immediate punish that makes the sleep turns losing, or the Rest merely delays a bad route without changing it.

- Example evidence:
 - local source: `engine/battle/effect_commands.asm` fails Rest before status

cure when HP is full.

- 200-game state regression: random play clicked full-HP Rest 798 times,

including 350 while statused; the filtered policy clicked it 0 times.

- long-battle drill 5: below-full Rest was useful in the Snorlax routes, but

only when the following sleep turns did not donate a decisive response.

- filtered seed 1 turn 12: toxic Raikou used below-full Rest but was KOed by

Snorlax Double-Edge before the recovery line could matter.

- filtered seed 16 turn 16: toxic Snorlax used below-full Rest but Skarmory

Whirlwinded it out, so the status cure had to be weighed against the forced Spikes entry.

- `mechanics_snorlax_full_hp_rest_status_fail`: full-HP paralysis did not make Rest legal; the scorer and strict labels now treat it as a failed mechanics line.

• `red_snorlax_vs_alakazam_curse_or_body_slam`: 84% unstatused Rest is too early into faster Recover/Psychic pressure; Body Slam is the live route.

- Implementation target: hard fail at exact full HP; separate scoring for

below-full status-cure Rest, emergency healing Rest, and tempo-losing Rest.

Setup Must Change The Damage Race

- Status: stable
- Type: sequence policy
- Rule: Setup is good when it changes the next-turn damage race, creates a KO

window, or leaves the boss ahead after the expected punish. Do not click setup merely because the boss is a setup-style Pokemon.

- Applies when: the boss survives, the setup improves KO math, or the player is unlikely to punish immediately.
- Does not apply when: the player has revealed lethal pressure, setup does not change the number of turns to win, or the boss loses too much tempo.

- Example fixtures:

- `bugsy_scyther_vs_quilava_fire_pressure`
- `clair_kingdra_vs_lapras_dragon_dance_or_attack`
- `whitney_clefairy_vs_bayleef_encore_window`
- `erika_victreebel_vs_snorlax_sleep_or_boost`
- `lance_yanma_vs_lapras_sleep_powder_or_quiver_dance`
- `will_slowbro_vs_houndoom_amnesia_or_surf`
- `red_charizard_vs_starmie_sunny_day_or_switch`

- Implementation target: plan scoring, then ROM setup weights.

Defensive Setup Still Needs Tempo

- Status: developing
- Type: sequence policy
- Rule: A defensive setup move is bad if the user is slower and already in

lethal range before the boost applies, even when the move boosts the correct defensive stat.

- Applies when: the player has revealed or highly likely faster damage that KOs, and the setup move does not act before that damage.
- Does not apply when: the boss moves first, survives the public hit, has sleep/paralysis/protection support, or the setup turn is forced by the opponent's inability to punish.
- Example fixtures:
 - `will_slowbro_vs_houndoom_amnesia_or_surf`
 - `pryce_piloswine_vs_typhlosion_amnesia_or_attack`
- Implementation target: setup scoring should check speed order and current KO range before rewarding the stat-stage improvement.

Two-Turn Packages Need A Safe First Turn

- Status: developing
- Type: sequence policy

- Rule: Do not choose a two-turn plan when the first turn loses to a revealed

immediate punish, unless the plan's first turn itself prevents that punish.

- Applies when: a line needs setup weather, a charge move, lock-in ramping,

recovery before attacking, or a switch before attacking, and the opponent has public damage or status that can stop the package before payoff.

- Does not apply when: the first turn changes the opponent's damage enough to

survive, the opponent is forced out, the user moves first and creates a forced state, or the line is explicitly a prediction from a losing position.

- Example fixtures:

- `red_charizard_vs_starmie_sunny_day_or_switch`

- Implementation target: plan scoring should evaluate the first-turn survival

and the second-turn payoff as a linked branch, not as separate move scores.

Explosion Needs A Forced Or Costly Target

- Status: developing
- Type: sacrifice policy
- Rule: Explosion is a cash-out only when the target cannot cheaply deny it, or

when the denial branch is itself punished by the resulting position.

- Applies when: the opponent has Protect, Ghost typing, a Normal-resistant pivot, a low-value sacrifice, or a clear switch that makes the Explosion trade bad.
- Does not apply when: the target is trapped, asleep with no safe wake branch,

forced to stay, too important to sacrifice, or the free entry after Explosion is itself the named route.

- Example evidence:
 - long-battle drill 2: Steelix Explosion into poisoned Snorlax was useful

because Spikes and follow-up attackers converted the chip into removal of the central threat.

- long-battle drill 2: Exeggutor Explosion into Misdreavus Protect did

nothing because the target denied the trade for free.

- Implementation target: sacrifice scoring should consider Protect, Ghosts,

Normal resists, forced-target evidence, and the next-entry plan.

Trap Clocks Override Normal Plans

- Status: developing
- Type: endgame policy
- Rule: Once Mean Look / Spider Web plus Perish Song or a similar forced clock

is active, evaluate the position as a countdown instead of a normal damage, Rest, or setup race.

- Applies when: the active Pokemon is trapped or likely to be trapped and a

countdown, perish, lock, or forced-faint route is available.

- Does not apply when: the trapped Pokemon can immediately KO, phaze, pivot

through Baton Pass-style mechanics, or accept the trade into a clearly winning endgame.

- Example evidence:
 - long-battle drill 2: Snorlax used Curse/Rest/Sleep Talk while Misdreavus

executed Mean Look plus Perish Song, causing both to faint on the clock.

- Smogon Misdreavus and SleepTrap resources describe Mean Look / Perish Song

as route-ending pressure that makes phazing, Pursuit, and sleep state central counterplay rather than side details.

- Benchmark artifact: `fixture_morty_misdreavus_vs_typhlosion_perish_route_001`;

mutation `mut_morty_perish_route_cannot_survive_001` flips to Gengar when Misdreavus no longer survives the public Flame Wheel branch long enough to start Perish Song.

- Implementation target: plan scoring should add a high-priority clock state

that asks "how many turns remain, and what action changes the clock?"

Do Not Collapse To One Route Blindly

- Status: developing
- Type: endgame policy
- Rule: Before sacrificing or exhausting the second-to-last useful route, name

the opponent's preserved anti-route and verify the last route can beat it.

- Applies when: a sacrifice, Explosion, Rest loop, hazard trade, or wall-war

commitment would leave only one realistic win condition.

- Does not apply when: the remaining route has a forced win, the opponent's

anti-route is gone or disabled, or the sacrifice immediately ends the game.

- Example evidence:
 - long-battle drill 3: Exeggutor's Explosion helped Snorlax beat Skarmory,

but left only Snorlax into a preserved Misdreavus Perish Song endgame.

- Implementation target: endgame scoring should count remaining independent routes, not only current matchup progress.

Prediction Needs A Base Case

- Status: developing
- Type: prediction policy
- Rule: A prediction-only move is good only if its miss branch remains

acceptable or the current board gives strong public evidence that the target will choose the predicted line.

- Applies when: Pursuit, double-switching, trap moves, immunities, Protect

reads, or choice-lock reads gain large value only if the opponent switches or chooses one specific move.

- Does not apply when: the prediction move is also the best or near-best move

if the opponent stays, or the board is losing unless the prediction succeeds.

- Example evidence:
 - Karen Houndoom into Alakazam: Pursuit is only superior if Alakazam

switches; Crunch is much better if Alakazam stays.

- Implementation target: prediction scoring should compare stay and switch

branches explicitly instead of rewarding "smart-looking" reads by default.

Reactive Counters Need Survival And Category

- Status: developing
- Type: reactive policy
- Rule: Counter, Mirror Coat, Destiny Bond, and similar reactive moves must

pass two checks before being scored as real answers: the user survives or otherwise acts in the right window, and the opponent's likely move belongs to the category or trigger the reactive move answers.

- Applies when: the move only pays off after taking a hit, after moving last,

or after the opponent chooses a specific damage category.

- Does not apply when: the reactive move is being used from a forced-losing

position where any normal move also fails and the gamble is the only route.

- Example evidence:
 - Red Blastoise into Zapdos: Mirror Coat is high-upside only if Blastoise

survives the Electric special hit.

- Red Blastoise into Tyranitar: Mirror Coat is poorly aligned because

Tyranitar's public punish profile includes Rock Slide and Earthquake.

- Implementation target: reactive scoring should require survival ranges,

category fit, and branch label before giving style or counterplay credit.

Sleep Is Strong But Must Be Gated

- Status: stable
- Type: hard rule
- Rule: Sleep is legal and strong; do not avoid it just because it feels mean.

Gate it only on real public fail states such as Sleep Clause, an already statused target, Substitute, Safeguard, or repeating sleep after it already landed. A landed sleep can justify one setup turn, then the boss should cash damage and only reapply sleep after the target wakes and Sleep Clause allows it.

- Applies when: target is awake, legal to sleep, and the miss risk is worth the

control.

- Does not apply when: the boss can simply KO now, Sleep Clause blocks it, or

missing loses a decisive race.

- Simulation note: in a Gen 2 Victreebel vs Snorlax scratch test, Sleep Powder

into one Swords Dance beat both raw attacking and raw setup over 200 seeds. Sleep Powder into two Swords Dances was worse than one boost, so the rule is "sleep buys the needed setup turn", not "sleep means boost forever".

- Example fixtures:
 - `morty_haunter_vs_noctowl_sleep_line`
 - `erika_victreebel_vs_snorlax_sleep_or_boost`
 - `lance_yanma_vs_lapras_sleep_powder_or_quiver_dance`
- Implementation target: sleep fail gates and status scoring.

One Debuff Can Be Right, Repeating Debuffs Usually Is Not

- Status: stable
- Type: sequence policy
- Rule: One accuracy or control debuff can be good when direct damage is nearly

worthless. Repeating debuffs is usually bad because the player can switch and clear them.

- Applies when: chip damage is negligible and one debuff materially improves

survival or tempo.

- Does not apply when: useful damage is available, the target can freely switch,

or the boss is spending multiple turns for no lasting gain.

- Example fixtures:
 - `falkner_pidgeotto_vs_geodude_rock_risk`
- Implementation target: sequence memory and repeat-utility penalties.

Hazards Need A Playable Setter Turn

- Status: developing
- Type: hazard policy
- Rule: Spikes are high-value long-game pressure, especially with this

romhack's three layers, but do not place hazards when the setter is slower and faces a revealed immediate KO unless the sacrifice creates a named route worth more than preserving the setter.

- Applies when: the active hazard setter is under public lethal pressure and

the expected hazard turn is not protected by sleep, paralysis, forced switch, resistance, or a known opponent setup/recovery turn.

- Does not apply when: the setter survives the public hit, the opponent is

likely forced out, the hazard layer creates an immediate forced route, or the setter is intentionally being spent as part of a cash-out plan.

- Example fixtures:
 - `koga_ariados_vs_typhlosion_spikes_or_toxic`
- Implementation target: hazard scoring should check speed order, public KO

ranges, and whether the layer's future value is actually reachable.

Local Spikes Must Be Layer-Aware

- Status: developing
- Type: romhack hazard policy
- Rule: Score Spikes from the current stack, not from a generic hazard slogan.

The third layer can be a high-value conversion turn when the player is grounded and hazard control is not public; a fourth layer is a failed move.

- Applies when: the local field state records player-side Spikes layers and

the boss is choosing whether to set Spikes, attack, switch, or sacrifice.

- Does not apply when: the setter faces immediate lethal pressure, the player

has an available spinner that can remove the stack cheaply, the player team is mostly Flying, or Explosion removes a more important defensive answer than the hazard layer would pressure.

- Example fixtures:

- `janine_qwilfish_finish_third_spikes_layer`
- `janine_qwilfish_spikes_already_maxed`

- Implementation target: action scoring and features should expose

`spikes_third_layer_available` and `spikes_already_maxed`, and the final decision should still review Explosion as a route trade.

Hazard Sacrifice Must Beat Hazard Control

- Status: developing
- Type: hazard policy
- Rule: Do not spend the Spiker just because a layer is down if the opponent's

spinner or remover survives and can clear it before the layer creates real pressure.

- Applies when: the Spiker is considering Explosion, sacrifice, or a low-value

trade while the opposing spinner, Defog-equivalent, or romhack Rapid Spin user is still active or easily preserved.

- Does not apply when: the sacrifice removes the hazard-control piece, traps or

punishes the remover, creates an immediate forced win, or the hazard has already produced the needed damage threshold.

- Example evidence:

- filtered long-battle drill 4: Cloyster set Spikes, Exploded into

Forretress, Forretress survived, spun the Spikes away, then traded for Raikou. The hazard plan did not survive the trade.

- Implementation target: hazard scoring should value "layer plus control" more

than "layer placed once".

Avoid Bad Lock-In Openers

- Status: stable
- Type: hard rule
- Rule: Do not begin a ramp or lock-in move like Rollout or Outrage when it is

resisted, does not KO, and gives the player a clear punish or switch.

- Applies when: the lock-in move starts from a weak position.
- Does not apply when: the target is already in KO range, paralyzed, unable to

punish, or the lock sequence is already safely underway.

- Example fixtures:
 - `whitney_miltank_vs_geodude_rollout_temptation`
- Implementation target: lock-move opener scoring.

Do Not Lock Into A Public Immunity

- Status: stable
- Type: hard rule
- Rule: Do not choose a Choice-locked identity STAB when public typing makes

the move fail. A live coverage move or preservation switch is better than committing to a no-effect move.

- Applies when: the target's public type or revealed mechanic makes the move

immune/no-effect and the boss item or move lock makes the mistake compound across future turns.

- Does not apply when: the target is expected to switch and the chosen move

covers the switch branch, or a local mechanic has been verified to bypass the immunity.

- Example fixtures:
 - `sabrina_alakazam_choice_lock_vs_houndoom`
- Implementation target: `public_type_immunity_risk` should fire from public

fixture text before generic coverage or identity bonuses can save the move.

Switch When Preservation Is Worth The Free Hit

- Status: developing
- Type: switch policy
- Rule: Switching is good when preserving the active boss Pokemon is worth more

than the free turn the player gets, especially if the bench has a clear resist or a better threat.

- Applies when: the current active is likely to lose or has high later value.
- Does not apply when: the switch-in is not actually safer, the active can take

a clean KO now, or the switch gives up too much pressure.

- Example fixtures:
 - `chuck_poliwrath_vs_pidgeotto_ice_punch`
 - `pryce_slowking_vs_ampharos_ground_pivot`
 - `brock_golem_vs_vaporeon_explosion_question`
 - `will_slowbro_vs_houndoom_amnesia_or_surf`
- Implementation target: switch preservation scoring and plan scoring.

Scout Hidden Coverage Before Committing An Ace

- Status: provisional
- Type: scout policy
- Rule: Hidden coverage suspicion can make probing, attacking once, or switching

better than greedy setup before committing an ace.

- Applies when: the player staying in only makes sense if they have dangerous

hidden coverage, the coverage branch is severe, and a public pivot preserves the route without handing over a worse position.

- Does not apply when: the threat is already revealed, no longer plausible, too

weak to change the route, or the boss has a direct winning line.

- Example fixtures:
 - `clair_dragonair_vs_suicune_hidden_ice_beam`
 - `lance_dragonite_vs_suicune_champion_ace`
- Benchmark artifact: `fixture_clair_dragonair_vs_suicune_hidden_ice_001`;

mutation `mut_clair_hidden_ice_absent_attack_now_001` flips to Thunder when the public hidden-Ice punish is no longer plausible.

- Implementation target: public hidden-coverage risk scoring.

Preserve Variety In Near-Tie Sacrifice Lines

- Status: developing
- Type: personality style
- Rule: If sacrifice, switch, and attack lines are genuinely close, the boss

should preserve some variety instead of becoming deterministic.

- Applies when: both lines are defensible and source notes call out near-tie

behavior.

- Does not apply when: one option is clearly forced by KO, public fail gate, or

survival.

- Example fixtures:
 - `brock_golem_vs_vaporeon_explosion_question`
 - `morty_gengar_vs_kadabra_destiny_bond`
- Implementation target: final tie-break/randomization policy.

Answered Teaching Notes

Erika Victreebel vs Snorlax

- Fixture: `erika_victreebel_vs_snorlax_sleep_or_boost`
- Answer: Sleep Powder first. Even though it is inaccurate, landing sleep gives

Victreebel the turn it needs to Swords Dance. After that, pressure with Sludge Bomb. If Snorlax wakes up, use Sleep Powder again only when Sleep Clause is legal.

- Extracted rule: Sleep can be the correct first move when it creates a setup

turn that changes the damage race. The follow-up is bounded: setup once, attack, and re-sleep only after a legal wake state.

Lance Yanma vs Lapras

- Fixture: `lance_yanma_vs_lapras_sleep_powder_or_quiver_dance`
- Answer: Same pattern as Erika. Sleep Powder first, then use the sleep turn to

Quiver Dance, then attack with Giga Drain. Reapply Sleep Powder only after Lapras wakes up and only if Sleep Clause is legal.

- Extracted rule: Weak immediate damage plus dangerous hidden coverage can still

favor sleep-first setup. The setup is not raw greed because sleep creates the turn that makes Quiver Dance safe enough to try.

- Implementation follow-up: the first scorer regression miss was not a new

strategic exception. `Quiver Dance` was missing from the setup taxonomy, so the debugger misclassified it as damage-like and awarded a false coverage bonus from the phrase "Ice coverage." Classifying `Quiver Dance` as setup now makes the debugger rank Sleep Powder first in this fixture.

- Direct-action follow-up: the Erika one-turn scorer now has the same

sleep-first opener logic as the multi-turn plan. In sleep/setup fixtures with an unstated target, a sleep move can earn `sleep_enables_setup_line` because it creates the turn that makes the boost legal. This is intentionally narrower than "sleep is always best."

- Plan follow-up: the next gap was sequence-level. Separate cards for

`Sleep Powder -> Giga Drain` and `Quiver Dance -> Giga Drain` still miss the actual taught line. The plan generator now emits a bounded `sleep_then_setup_then_attack` card for fixtures tagged both `sleep` and `setup`, with branches that stop and re-score after miss, wake, switch, Sleep Clause, or lost setup value.

External GSC Rhydon vs Sleeping Snorlax

- Fixture: `external_gsc_rhydon_vs_sleeping_snorlax_curse_window`
- Source: replay 912130, turn 49.
- Answer: Curse is better than immediate Earthquake, Rock Slide, or switching

Zapdos in this exact endgame because Snorlax is still publicly asleep and Rhydon needs the boost to convert the route before or through the wake branch. The line is not "setup because setup is good"; it is setup because the sleep window changes the KO math and still leaves a coherent turn after Snorlax wakes.

- Extracted rule: a sleeping target can create a real setup window, but the

explanation must still price wake timing, speed changes, entry damage, and the next attack. Preserve the phrase "sleeping win condition is not available yet" as a state-tracking warning, not as a reason to ignore the target.

- Implementation follow-up: the scorer now has a narrow

`sleeping_target_setup_window` contribution for setup moves when the public target is asleep in a fixture tagged both `sleep` and `setup`. This fixed the new replay-derived fixture without globally rewarding setup.

Mechanics Drill: Full-HP Rest And Awake Sleep Talk

- Fixture: `mechanics_snorlax_full_hp_rest_status_fail`
- Answer: do not use Rest at full HP to cure paralysis, and do not use Sleep

Talk while awake. Body Slam or even a phazeable Curse line is better than a known failed move.

- Extracted rule: Rest status-cure logic only matters after the move passes the

full-HP check. If the user is at exactly full HP, status does not rescue the move. Sleep Talk similarly requires actual sleep, not merely any status.

- Implementation follow-up: the debugger now applies `full_hp_rest_fails`,

`healthy_rest_no_status`, and `awake_sleep_talk_fails` contributions, and the feature report exposes `rest_at_full_hp`, `rest_high_hp_no_status`, and `sleep_talk_while_away` features.

Janine Qwilfish Romhack Spikes Layers

- Fixtures: `janine_qwilfish_finish_third_spikes_layer` and

`janine_qwilfish_spikes_already_maxed`

- Answer: at two local layers, Qwilfish should usually finish the third layer

over ordinary attack chip or switching when the player is grounded, no player spinner is public, and Qwilfish is not under immediate lethal pressure. At three layers, another Spikes click is a known failed move and should be rejected before style or long-game reasoning.

- Extracted rule: local Spikes decisions require the current layer count. The

third layer changes future grounded switch-ins from max HP / 6 to max HP / 4; a fourth layer is not "more pressure", it is no action. Explosion stays in a separate route-trade bucket because removing Snorlax or a wall can be worth more than completing the stack in some states.

- Implementation follow-up: the debugger now applies

`third_spikes_layer_pressure` and `spikes_already_maxed` contributions, and the feature report exposes `move_class_hazard`, `spikes_third_layer_available`, and `spikes_already_maxed`.

Current Scorer Gap Audit

Most entries below are notebook interpretations of the existing queue, user feedback, and local damage debugger. Treat them as review targets until direct action labels or stronger trajectory rows confirm them; entries marked resolved have already been promoted into fixtures, labels, or code changes.

Latest local audit status:

- `python -m tools.boss_ai_preference validate` passes and now validates 30

public state-transition cards and 30 hidden oracles with explicit `seed` / `holdout` / `fixture_harvest` splits.

- `python -m tools.boss_ai_preference benchmark-policy` writes

`audit/boss_ai_preference/state_transition_policy_answers.json` from the deterministic `state_transition_baseline_v1` policy. This policy reads `state_transition_public_cards.json`, emits `state_hash`, `candidate_ranking`, inferred catastrophe branches, answer-changing information, and `rules_fired`, and does not read hidden oracle labels.

- ``python -m tools.boss_ai_preference benchmark-report --answers`

`audit/boss_ai_preference/state_transition_policy_answers.json` writes the executable state-transition benchmark report by joining the hidden oracle file only inside the evaluator. The current report has `benchmark_contract_ready=True`, `policy_evaluated=True`, `policy_passes=True`, and `split_passes={'fixture_harvest': True, 'holdout': True, 'seed': True}` across the 30 current snapshots. This is a baseline gate, not mastery proof.

- `python -m tools.boss_ai_preference benchmark-harvest` writes

`audit/boss_ai_preference/fixture_benchmark_harvest.md`. The current harvest finds 4 complete fixture-derived benchmark candidates: `chuck_poliwrath_vs_pidgeotto_ice_punch`, `janine_qwilfish_finish_third_spikes_layer`, and `mechanics_snorlax_full_hp_rest_status_fail`, and `brock_golem_vs_vaporeon_explosion_question`. Those four are now promoted into the `fixture_harvest` benchmark split. The Will Slowbro/Houndoom fast-threat case is also promoted from high-confidence pairwise feedback as a targeted coverage card with a pivot-unavailable mutation. The Koga Ariados/Typhlosion case is also promoted to test that Spikes/status lose to a revealed lethal Fire punish, the Bugsy Scyther/Geodude case is promoted to test that setup can be correct when the public punish is survivable, and the Chuck Poliwrath/Alakazam case is promoted to test that a verified Dark pivot can dominate an accuracy-dependent sleep line. The Morty Haunter/Noctowl case is promoted to test that legal sleep can dominate fixed damage and premature ace exposure. The Falkner Pidgeotto/Geodude case is promoted to test that a single scout probe can dominate tiny direct chip when a public punish branch is live. The other 23 feedback fixtures are partial, mostly because they lack an acceptable alternative label.

- `python -m tools.boss_ai_preference benchmark-label-queue` writes

`audit/boss_ai_preference/benchmark_label_queue.md`. The current queue has 23 review requests and 11 one-label completions where an unclassified action can be reviewed as the missing acceptable alternative. It prefers plausible probes such as preservation switches or sensible coverage over already catastrophic moves, so the queue is for evidence collection rather than label invention. The remaining partials need either a contextual exception, a new benchmark variant, or a resolved single-best / catastrophe label.

- `python -m unittest tools.boss_ai_preference.tests.test_benchmark_positions`

also checks answer-changing mutations: blocked Sleep Clause changes away from sleep, maxed Spikes changes away from Spikes, revealed Protect changes away from Explosion, unavailable Piloswine changes away from the preservation switch, unavailable Sudowoodo changes from preservation to Ice Punch, unavailable Houndoom changes from preservation to Surf, unavailable Tentacruel changes the Koga Fire-pressure preservation pivot to Umbreon, and a removed safe setup window changes Bugsy Scyther from Swords Dance to Wing Attack. Unavailable Umbreon changes Chuck Poliwrath from preservation to Hypnosis. Occupied Sleep Clause changes

Morty Hunter from Hypnosis to Night Shade. Removing the public Rock-risk branch changes Falkner Pidgeotto from Sand Attack to Gust. Full-HP Rest / awake Sleep Talk remain failed state transitions. It also checks replay-derived opening, Explosion, RestTalk, and late Rapid Spin boundaries from Showdown GSC logs.

- `python tools/audit/check_boss_ai_preference.py` passes.
- `python tools/audit/check_boss_ai_trace_invariants.py` passes.
- `python tools/audit/check_boss_ai_policy_contract.py` passes.
- `python tools/audit/check_boss_ai_no_cheat.py` passes.
- `python tools/audit/check_boss_ai_gating.py` passes.
- `python tools/audit/check_boss_ai_memory_budget.py` passes.
- ``python -m tools.boss_ai_debugger regress --json-out`

`audit/boss_ai_preference/regression_report.json` now reports 43 / 43 strict pairwise labels agreeing with the scorer after the Quiver Dance` taxonomy fix, the sleeping-target setup-window rule, the Rest/Sleep Talk state gates, the local Spikes layer-state rules, and the sleep-first setup opener rule.

- Resolved strict regression:

`lance_yanma_vs_lapras_sleep_powder_or_quiver_dance`. Before the fix, the debugger ranked Quiver Dance 59 over Sleep Powder 58 because `Quiver Dance` was absent from `SETUP_NAMES` / `SETUP_MOVES`, so the word "coverage" in "Ice coverage" triggered the damage coverage bonus. After the fix, the same inspection ranks Sleep Powder 58, Giga Drain 57, Switch to Kingdra 56, and Quiver Dance 48 with `setup_identity` and `setup_risk` applied.

- Resolved plan-card gap: `audit/boss_ai_preference/plan_queue.md` and

`audit/boss_ai_preference/coach_report.md` now surface `sleep_then_setup_then_attack` for both the Lance Yanma and Erika Victreebel sleep/setup demonstrations. This does not prove the plan is always correct; it means the review tool can finally show the taught candidate instead of forcing reviewers to choose among incomplete fragments.

- Resolved replay-fixture gap:

`external_gsc_rhydon_vs_sleeping_snorlax_curse_window` converts the 54-turn external review into a scored single-position test. The debugger now ranks Curse 68, Earthquake 63, Rock Slide 57, and Switch to Zapdos 54, with the setup score coming from public sleep state rather than vague setup prose.

- Resolved Rest/Sleep Talk state-gate gap:

`mechanics_snorlax_full_hp_rest_status_fail` converts the local Rest source check into a scored mechanics fixture. The debugger now ranks Curse 56, Body Slam 50, Rest 39, and Sleep Talk 38, explicitly penalizing full-HP Rest and awake Sleep Talk instead of treating them as generic recovery/control.

- Resolved romhack Spikes layer gap:

`janine_qwilfish_finish_third_spikes_layer` and `janine_qwilfish_spikes_already_maxed` convert the three-layer Spikes source check into scored mechanics fixtures. The debugger now ranks third-layer Spikes above ordinary chip in the safe Janine Qwilfish fixture, and ranks maxed-out Spikes last once the player side already has three layers.

- Resolved Erika direct-action gap:

`erika_victreebel_vs_snorlax_sleep_or_boost` now has strict pairwise labels for Sleep Powder over raw Swords Dance, Sludge Bomb, and switching. The debugger ranks Sleep Powder 71 over Swords Dance 68 because the sleep move is scored as the opener to a setup line rather than generic status flavor.

- Resolved Will preservation gap:

`will_slowbro_vs_houndoom_amnesia_or_surf` now has strict pairwise labels for switching to Will's Houndoom over Amnesia, Surf, and Psychic. The lesson is preservation under a faster public KO branch: boosting the right defense and clicking super-effective Surf both lose if Slowbro faints before moving.

- Resolved Koga hazard-passivity gap:

`koga_ariados_vs_typhlosion_spikes_or_toxic` now has trajectory evidence and an executable benchmark showing that Spikes and Toxic are too slow when revealed Flamethrower is a guaranteed KO into Ariados. The taught transition is Tentacruel first, re-score next; the mutation removes Tentacruel and flips the answer to Umbreon rather than pretending hazards are still free.

- Resolved Buggy safe-setup gap:

`bugsy_scyther_vs_geodude_safe_swords_dance` now has trajectory evidence and an executable benchmark showing the positive side of setup arbitration. Swords Dance is preferred because Scyther is faster, direct Wing Attack is low-impact, and the public Rock Throw branch does not remove Scyther from full HP. The mutation lowers Scyther into a KO branch and flips the answer to Wing Attack.

- Resolved Buggy fire-setup gap:

`bugsy_scyther_vs_quilava_fire_pressure` now has an executable benchmark grounded in saved pairwise and trajectory feedback. Swords Dance is preferred because Scyther acts first, survives one Ember, unboosted Wing Attack does not KO, and boosted Wing Attack changes the two-turn route. The mutation lowers Scyther into an Ember KO branch and flips the answer to Wing Attack.

- Resolved Chuck Psychic-pivot gap:

`chuck_poliwrath_vsalakazam_hypnosis_or_ice_punch` now has trajectory evidence and an executable benchmark showing that Hypnosis is a real fallback but not the best line while Umbreon is healthy. The romhack type chart keeps Psychic at no effect into Dark, so switching to Umbreon preserves Poliwrath from faster Psychic/Recover pressure; removing Umbreon flips the mutation to Hypnosis.

- Resolved Morty sleep-control gap:

`morty_haunter_vs_noctowl_sleep_line` now has trajectory evidence and an executable benchmark showing the positive sleep-control case. Hypnosis is preferred because Sleep Clause is free, Noctowl is unstated, and the public Peck branch is survivable; Night Shade becomes the fallback when Sleep Clause is occupied.

- Resolved Falkner scout-probe gap:

`falkner_pidgeotto_vs_geodude_rock_risk` now has an executable benchmark grounded in existing pairwise and trajectory feedback. One Sand Attack is preferred because Gust is tiny chip while public Rock Throw is a meaningful punish. The benchmark also records the exception: repeated probing is not the policy, and if the Rock-risk branch is removed the mutation flips to Gust.

- Resolved Pryce fire-pivot gap:

`pryce_cloyster_vs_quilava_explosion_line` now has an executable benchmark grounded in saved pairwise feedback and Smogon bad-matchup switching / Explosion discipline principles. Slowking is preferred because revealed Flame Wheel can remove 39% Cloyster before Surf or Explosion makes progress, while Cloyster still has future Spikes/Explosion value. The mutation raises Cloyster out of the KO branch and flips the answer to Surf.

- Resolved Whitney lock-sequencing gap:

`whitney_miltank_vs_geodude_rollout_temptation` now has an executable benchmark grounded in saved pairwise and trajectory feedback. Body Slam is preferred before Rollout because the first Rollout hit is tiny, Geodude is not paralyzed yet, and Miltank is healthy enough to pressure before healing. The mutation marks Geodude paralyzed and flips the answer to Rollout.

- Resolved Pryce preservation gap:

`pryce_slowking_vs_ampharos_ground_pivot` now has strict pairwise labels for switching to Piloswine over Thunder Wave, Surf, and Rest. The important romhack-specific detail is that Dragon's Majesty means Piloswine is not fully immune to Ampharos's Electric damage, but it still resists the pressure, preserves Slowking, and threatens back with Ground coverage.

- Resolved Buggy Ariados status-tempo gap:

`bugsy_ariados_vs_pidgey_toxic_or_attack` now has strict pairwise labels for Toxic over Giga Drain, Leech Life, Poison Sting, and switching to Scyther. The lesson is not "status because status"; Toxic is preferred because Pidgey's Flying pressure is real but not lethal, Ariados's chip is weak, and the poison clock changes future turns without exposing the ace early.

- Resolved Sabrina public-immunity gap:

`sabrina_alakazam_choice_lock_vs_houndoom` now penalizes Psychic with `public_type_immunity_risk` when public text says the target is a Dark type, and strict labels prefer Hypno, ThunderPunch, and Ice Punch over the failed Choice Specs Psychic lock.

Falkner Pidgeotto vs Geodude

- Fixture: `falkner_pidgeotto_vs_geodude_rock_risk`

- Current lesson: one Sand Attack is a real scout/probe turn because Gust is nearly meaningless and public Rock Throw can swing the game immediately.
- Damage evidence:
 - Pidgeotto Gust into 86% Geodude is 2-3 damage, 6-9% of current HP.
 - Geodude Rock Throw into 72% Pidgeotto is 27-32 damage, 96-114% of current HP.

- Geodude Tackle into Pidgeotto is 7-9 damage, 25-32% of current HP.
- Failure signs:
 - repeating Sand Attack several times without a conversion plan;
 - switching only to avoid risk when Noctowl still takes heavy damage;
 - clicking Gust and calling it pressure despite doing under 10%.

Pryce Cloyster vs Quilava

- Fixture: `pryce_cloyster_vs_quilava_explosion_line`
- Current lesson: low HP does not automatically make Cloyster expendable.

If the public faster Flame Wheel branch removes Cloyster before Surf or Explosion resolves, the right transition is Slowking first, preserving Cloyster's later Spikes/Explosion route.

- Boundary mutation:
 - when Cloyster is raised out of the Flame Wheel KO branch, the policy flips

from `switch_slowking` to `move_surf`;

- Explosion still needs a named route and speed/survival evidence.
- Failure signs:
 - saying "Surf is super-effective" without checking whether Cloyster moves;
 - saying "Cloyster is low, so boom" without naming the route opened;
 - using Protect as a loop instead of a one-turn scout before re-scoring.

Bugsy Scyther vs Quilava

- Fixture: `bugsy_scyther_vs_quilava_fire_pressure`
- Current lesson: setup is conditional on speed, survival, and conversion, not

on the fact that Scyther is an ace. The saved feedback prefers Swords Dance over raw Wing Attack when Scyther acts first and survives one Ember, but the damage range is close enough that this is now an executable boundary benchmark.

- Damage evidence:

- unboosted Wing Attack into 69% Quilava is 21-25 damage, 49-58% of current

HP, never a KO;

- Quilava Ember into 81% Scyther is 32-38 damage, 71-84% of current HP, never

a non-crit KO;

- +2 Wing Attack using doubled attack stat is 41-49 damage, 95-114% of

current HP, a possible but not guaranteed KO.

- Failure signs:
 - treating "setup once then attack" as automatic without checking speed and

the low-roll branch;

- cashing weak unboosted damage when it leaves Quilava alive anyway;
- ignoring that a crit Ember invalidates the line but should be separated as

a luck branch, not necessarily a strategic error.

- Benchmark artifact: `fixture_bugsy_scyther_vs_quilava_fire_setup_001`;

mutation `mut_bugsy_fire_setup_window_removed_001` flips to Wing Attack when Ember removes the setup route before boosted Wing Attack can convert.

Whitney Miltank vs Geodude

- Fixture: `whitney_miltank_vs_geodude_rollout_temptation`
- Current lesson: Body Slam is the opener because first-turn Rollout is weak

by public damage evidence before it ramps. The taught plan is Body Slam pressure, heal with Milk Drink around the danger threshold, and only consider Rollout after paralysis or a safer setup state.

- Damage evidence:
 - Body Slam into 78% Geodude is 7-9 damage, 15-19% of current HP;
 - first-turn Rollout into 78% Geodude is 2-3 damage, 4-6% of current HP;
 - Geodude Rock Throw into 92% Miltank is 15-18 damage, 21-25% of current HP.
- Failure signs:
 - opening Rollout because it is iconic;
 - using Milk Drink at high HP as a scout when Body Slam has not created

paralysis or meaningful chip yet;

- repeating Body Slam forever without noticing when the board becomes safe

enough to ramp.

- Benchmark artifact: `fixture_whitney_miltank_vs_geodude_rollout_lock_001`;

mutation `mut_whitney_rollout_after_status_001` flips to Rollout once paralysis and board safety make the lock route live.

Chuck Poliwrath vs Pidgeotto

- Fixture: `chuck_poliwrath_vs_pidgeotto_ice_punch`
- Current lesson: Ice Punch is the best listed attack, but the saved feedback

says the better plan is switching to Sudowoodo because Sudowoodo walls the public Flying/Normal pressure and preserves Poliwrath's HP. Focus Punch is not reliable into a faster attacker that is likely to hit.

- Damage evidence:
 - Poliwrath Ice Punch into 58% Pidgeotto is 34-40 damage, 52-61% of current

HP, never a non-crit KO;

- Poliwrath Focus Punch into 58% Pidgeotto is 58-69 damage, 88-105% of

current HP if it resolves, but it is vulnerable to interruption;

- Pidgeotto Wing Attack into 76% Poliwrath is 52-62 damage, 53-63% of

current HP;

- Pidgeotto Wing Attack into 58% Sudowoodo is 11-14 damage, 17-22% of

current HP.

- Failure signs:
 - choosing Focus Punch because its damage is attractive if the opponent does

not attack;

- choosing Ice Punch without asking whether Poliwrath's HP is needed later;
- switching to any resist without naming the preserved job and the next turn.

Red Snorlax vs Alakazam

- Fixture: `red_snorlax_vs_alakazam_curse_or_body_slam`
- Current lesson: this is unresolved and should stay in the queue. Body Slam

immediately creates paralysis and chip chances; Curse improves future damage but still does not immediately beat Recover/Psychic pressure. Rest at 84% with no status is not a serious line.

- Damage evidence:

- unboosted Body Slam into 62% Alakazam is 43-51 damage, 28-34% of current

HP;

- +1 Body Slam into 62% Alakazam is 65-77 damage, 43-51% of current HP;
- Alakazam Psychic into 84% Snorlax is 133-157 damage, 47-55% of current HP.
- Failure signs:
 - choosing Curse solely because Snorlax is a setup wallbreaker;
 - choosing Rest as a generic recovery move before status or danger threshold;
 - ignoring Recover as the player's likely answer to slow chip.

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